Supplementary Material

A Two-Grid Characterization of Euler Sampling Accuracy in Flow Matching

Dev D. Goyal

This supplement supports the main article with complete proofs, explicit constants, and experimental methods. The table below shows where to find each part of the argument. Supplementary section, equation, theorem, figure, and table numbers start with S. References without S point to the main article. Each document starts at page one.

Main-article claim	Supplementary sections
Two-grid characterization and error lower bound	S1–S3. Uniform estimates and the proof.
Odd-dimensional nonexistence result	S4 and S9. Gaussian and radial-source arguments.
Initial-risk constraint	S5. Quadratic tests and explicit constants.
Exact examples and scalar optimum	S6, S10, and S11. Constructions and scope.
Quantitative residual bound and solver extensions	S7–S8. Coefficient bounds, spectral tail, and numerical comparisons.
Sample-based evaluation	S12–S14 and S16. Confidence bounds, network regularity, protocol, and variability across evaluation sets.
Learned scalar fields	S15. Training, quadrature, and results for each model.
Population mixture comparison	S14-H. Exact Gaussian block, quantile integration, and all resolutions.

S1. Uniform estimates and compactness

The fitted velocity may change when the number of steps changes. We therefore need estimates with the same constants for every velocity in the allowed class. This is what "uniform" means below. An estimate for one fixed velocity would not justify optimizing over the whole class. We derive these estimates from the numerical recursion itself, without a Wasserstein tangent expansion.[1]

All norms of spatial derivatives below are operator norms, which measure the largest output produced by an input of unit norm. Second derivatives are bilinear operator norms, with both input vectors restricted to unit norm. The time interval is $[0, T]$, with $0 < T \le 1$. Write $\gamma_d = \mathcal{N}(0, I_d)$. The class $\mathcal{B}_K$ and its C^1_{loc} closure $\overline{\mathcal{B}}_K$ are those of Eq. (1). The closure includes limits of sequences whose fields and first derivatives converge uniformly on each compact space–time set. We include both endpoints of the time interval and use one-sided time derivatives there. A limiting field need not have continuous second derivatives.

A. *Constants and deterministic estimates*

We first control how far particles move, how the exact flow depends on its starting point, and how Euler error accumulates. The constants depend only on the common derivative bound and source moments. They therefore remain valid when the fitted field changes. Set

$$C_F = e^{KT}, \qquad A = K(1+K), \qquad B = K(1+K)^2, \tag{S1}$$

$$J = B + KA, \qquad C_D = K + 2K^2, \qquad C_E = AC_F^3, \tag{S2}$$

$$M_2 = 2(1+d), \qquad M_4 = 8\{1 + d(d+2)\}, \qquad M_3 = M_4^{3/4}. \tag{S3}$$

These are conservative upper bounds. We use B only for this scalar constant; model-path bias is denoted $\mathfrak{b}(u)$. The variational equations describe the derivatives of the flow with respect to its starting point. Applying Gronwall's inequality to those equations and to the trajectories gives, for the flow $F_{s,t}$ of any $u \in \mathcal{B}_K$,

$$1 + |F_{s,t}(x)| \le e^{K|t-s|}(1 + |x|), \qquad \|DF_{s,t}\| \le e^{K|t-s|}, \tag{S4}$$

$$\|D^2 F_{s,t}\| \le K|t-s|e^{2K|t-s|}, \qquad |a_t(x)| \le A(1+|x|), \tag{S5}$$

$$\|Da_t(x)\| \le C_D(1+|x|), \qquad |j_t(x)| \le J(1+|x|)^2, \tag{S6}$$

where $a = \partial_t u + Du\, u$ is the acceleration along an exact trajectory and $j = \partial_t a + Da\, u$ is its time derivative along that trajectory, sometimes called jerk. The Jacobian $DF_{s,t}$ records how a small change in the starting point changes the final position. The Hessian records how this Jacobian changes. To obtain its bound, differentiate the Jacobian equation and integrate $\|D^2 u\|\|DF\|^2$ using the solution of the associated homogeneous linear equation. The next identity follows by expanding the derivative along a trajectory.

$$j - Du\, a = b := \partial_{tt} u + 2D\partial_t u\, u + D^2 u[u,u], \qquad |b(t,x)| \le B(1+|x|)^2 \tag{S7}$$

[1] AI assistance for this section is disclosed in the acknowledgment [1–3].

The right side contains exactly the second derivatives controlled by Eq. (1).

To compare individual trajectories, start the exact flow and Euler from the same random point $G \sim \gamma_d$. This pairs their outputs and thus defines a coupling of the two distributions. The root mean squared displacement under this pairing gives an upper bound on Wasserstein distance. The pairing need not be optimal. If $S_{2,j} = \sum_{k<j} h_k^2$, then

$$1 + |X_{t_j}|,\ 1 + |\widehat{X}_j| \le C_F(1 + |G|), \tag{S8}$$

$$|\widehat{X}_j - X_{t_j}| \le C_E S_{2,j}(1 + |G|). \tag{S9}$$

Here the particle defect means the difference between one Euler update and one exact update from the same position. It is at most $Ah^2 e^{Kh}(1 + |x|)/2$ and the Euler map has Lipschitz constant $1 + Kh$. The Lipschitz bound controls the propagation of errors from earlier steps. Combining the one-step defects with these propagation factors gives the displayed C_E bound. The moments satisfy $\mathbb{E}(1 + |G|)^2 \le M_2$, $\mathbb{E}(1 + |G|)^4 \le M_4$, and $\mathbb{E}(1 + |G|)^3 \le M_3$. These are moments of the Gaussian starting point, not moments of a target law.

The next lemma allows us to take limits of fitted fields. It also ensures that their exact and numerical distributions converge to those of the limiting field.

Lemma S1 (Compact closure and continuous laws)**.** *The class $\overline{\mathcal{B}}_K$ is sequentially compact in C^1_{loc}. Its elements are C^1, with locally Lipschitz first derivatives, globally K-Lipschitz spatial velocities, and the growth bounds in Eq. (1). They have complete C^1 bi-Lipschitz flows on the time strip. If $u_n \to u$ in C^1_{loc}, then the flows and their spatial Jacobians converge uniformly on compact cylinders, and*

$$\sup_{0 \le t \le T} \|F^{u_n}_{0,t}(G) - F^{u}_{0,t}(G)\|_{L^2} \longrightarrow 0.$$

The same L^2 continuity holds for the Euler law on any fixed finite grid.

Proof: We first prove convergence on bounded space–time regions. We then control the contributions from starting points outside those regions. A compact cylinder here is a compact time interval times a bounded closed spatial set. On each compact cylinder the first derivatives are bounded and uniformly Lipschitz, by the three second-derivative bounds in Eq. (1). The function values are bounded there by the growth bound. Arzelà–Ascoli, a diagonal subsequence, and the fundamental theorem of calculus identify a C^1 limit with locally Lipschitz first derivatives. The local topology can be described by a metric. Taking the closure therefore preserves this compactness property. The common linear-growth and spatial Lipschitz bounds give flows throughout the interval, including backward time. Gronwall's inequality on a common compact set of trajectories gives flow convergence. The Jacobian equation has coefficients Du_n that converge locally uniformly, so the same argument gives convergence of the Jacobians.

For $|G| \le R$, the convergence is uniform in time. On the complement, the square of the difference is at most $4C_F^2(1+|G|)^2$, an integrable Gaussian tail independent of n. Taking $R \to \infty$ proves the asserted L^2 convergence. On a fixed grid, Euler applies only finitely many update maps. These maps converge locally uniformly and satisfy the same growth bound, so the same argument applies to their composition. Bounds (S6) involving first derivatives pass directly to the limit. Second-derivative bounds hold almost everywhere. The Taylor and weak-error estimates below pass from smooth approximations because their remainders have common bounds. This argument does not require C^2 convergence. ∎

We also need to pass from accuracy at grid nodes to accuracy at every time. For a prescribed W_2-continuous path p starting at γ_d, write

$$\mathfrak{b}(u) = \sup_{t \in [0,T]} W_2(\rho^u_t, p_t), \qquad \omega_p(r) = \sup_{|t-s| \le r} W_2(p_t, p_s).$$

The first quantity is the largest continuous-model bias along the path. The second bounds how much the prescribed distribution can change over a short time interval. Compactness and the preceding lemma make $\mathfrak{b}$ continuous on the closure. With E_c defined at the coarse nodes, Eq. (S9) gives

$$\mathfrak{b}(u) \le E_c + C_E\sqrt{M_2}\,TH + KC_F\sqrt{M_2}\,H + \omega_p(H). \tag{S10}$$

The second term bounds integration error at nodes, and the final two terms compare a time with the preceding node. For the independent interpolation (9), $\omega_p(H) \le \sqrt{d+\nu}\,H$, where $\nu = \operatorname{tr} S$. Consequently

$$\mathfrak{b}(u) \le E_c + C_{\mathrm{path}}H, \qquad C_{\mathrm{path}} = C_E\sqrt{M_2}\,T + KC_F\sqrt{M_2} + \sqrt{d+\nu}. \tag{S11}$$

This bound needs no exact marginal preservation by u.

S2. Signed recursive errors and the two-grid contrast

A test average can change in either direction. We keep its sign so we can tell when changes from different steps add or cancel. To use the same measurement at different times, trace a point backward through the exact flow before applying the

starting test. The value is constant along each exact trajectory, so the exact-model average is shared by both runs. Subtracting their errors cancels that average.[2]

Let $\phi \in C^{1,1}(\mathbb{R}^d)$ with $\|\nabla\phi\|_\infty \le 1$ and $\mathrm{Lip}(\nabla\phi) \le 1$. Set

$$\phi_t = \phi \circ F_{t,0}, \qquad I_u(t) = \mathbb{E}[\nabla\phi_t(X_t) \cdot a_t(X_t)]. \tag{S12}$$

The exact-flow expectation $\mathbb{E}\phi_t(X_t) = \mathbb{E}\phi(G)$ is constant. The quantity I_u measures the effect of acceleration on this time-dependent test. The next constants control its variation and the remainders in the expectation error. Such an error is called a weak error because it compares averages of tests, rather than the positions of individual particles. Define

$$H_\phi = (1+KT)C_F^3, \tag{S13}$$
$$C_0 = JC_F^3/6 + KAC_F^2/2 + TH_\phi A^2 C_F^2/8, \tag{S14}$$
$$C_b = H_\phi A + C_F C_D, \tag{S15}$$
$$C_w = C_0 C_F^2 M_2 + \tfrac{1}{2} C_b C_F C_E M_2 T, \qquad M_w = BC_F^3\sqrt{M_4}. \tag{S16}$$

Lemma S2 (Uniform weak error, including subintervals)**.** *For every pair of nodes $i < j$, write $D_k = \mathbb{E}\phi_{t_k}(\widehat{X}_k) - \mathbb{E}\phi(G)$. Then*

$$D_j - D_i = -\tfrac{1}{2}\sum_{k=i}^{j-1} h_k^2 I_u(t_k) + R_{ij}, \qquad |R_{ij}| \le C_w H \sum_{k=i}^{j-1} h_k^2. \tag{S17}$$

Moreover $\mathrm{Lip}(\phi_t) \le C_F$ *and* $\mathrm{Lip}(I_u) \le M_w$*, uniformly over fields in* $\overline{\mathcal{B}}_K$.

Proof: We first compare a numerical update and an exact update from the same point. We then include the error already present in the numerical starting point. This second comparison is needed because Euler does not restart from the exact distribution at each step. First take a smooth field. For $Y = F_{t,t+h}(x)$ and $E = x + hu(t,x)$, trajectory Taylor expansion gives

$$E - Y = -\tfrac{1}{2}h^2 a_t(x) + R, \qquad |R| \le JC_F^2 h^3 (1+|x|)^2/6,$$

and the lower-order estimate is $|E-Y| \le AC_F h^2(1+|x|)/2$. The transported test has Hessian bound H_ϕ, and

$$\nabla\phi_{t+h}(Y) = DF_{t,t+h}(x)^{-\top}\nabla\phi_t(x).$$

The difference from $\nabla\phi_t(x)$ is at most KhC_F. The $C^{1,1}$ Taylor inequality therefore proves

$$\phi_{t+h}(E) - \phi_t(x) = -\tfrac{1}{2}h^2 \nabla\phi_t(x)\cdot a_t(x) + R_0, \quad |R_0| \le C_0 h^3 (1+|x|)^2.$$

The three terms in C_0 bound the three remainder contributions in this one-step comparison. The spatial gradient of $\nabla\phi_t \cdot a_t$ has norm at most $C_b(1+|x|)$ almost everywhere. Comparing its numerical-state and exact-state expectations by (S9) costs at most $C_b C_F C_E M_2 S_{2,k} \le C_b C_F C_E M_2 TH$ at every node, including nodes after i. Now sum over the requested subinterval to obtain (S17). The common TH bound includes particle error accumulated before i. Thus this is an estimate for the actual recursive Euler computation, not for a computation restarted from exact marginals.

Let $J_t(x) = DF_{0,t}(x)$ and $U_t = J_t^{-1} a_t \circ F_{0,t}$. Equation (S7) gives $\partial_t U_t = J_t^{-1} b_t \circ F_{0,t}$ and $\|\partial_t U_t(G)\|_2 \le BC_F^3\sqrt{M_4}$. Since $I_u(t) = \mathbb{E}[\nabla\phi(G)\cdot U_t(G)]$, this proves the Lipschitz bound. Smooth approximations of the test preserve its first-derivative and Lipschitz bounds. To pass to the C^1 closure, use Lemma S1 and local convergence of accelerations on bounded sets. Gaussian moment bounds control the remaining tails. The time-Lipschitz bound passes as an inequality. It does not assert that a limiting second derivative is continuous. ∎

Let $\widetilde{X}_{2k}$ denote the recursive Euler state on the bisected grid at the common node t_k, for the *same* field. Define

$$Z_k = \mathbb{E}\phi_{t_k}(\widehat{X}_k) - \mathbb{E}\phi_{t_k}(\widetilde{X}_{2k}).$$

Lemma S3 (Two-grid contrast)**.** *For any $i < j$,*

$$Z_j - Z_i = -\tfrac{1}{4}\sum_{k=i}^{j-1} h_k^2 I_u(t_k) + \mathcal{R}_{ij}, \qquad |\mathcal{R}_{ij}| \le C_{\times,w} H \sum_{k=i}^{j-1} h_k^2, \quad C_{\times,w} = \tfrac{5}{4}C_w + M_w/16. \tag{S18}$$

Also $|Z_k| \le C_F(E_c + E_f)$, without assuming that $\rho_t^u = p_t$.

Proof: The fine-grid leading term contributed by one coarse interval is $-h_k^2\{I_u(t_k) + I_u(t_k + h_k/2)\}/8$. Subtract it from the coarse term $-h_k^2 I_u(t_k)/2$ to obtain

$$-h_k^2 I_u(t_k)/4 + h_k^2\{I_u(t_k + h_k/2) - I_u(t_k)\}/8.$$

The second summand is at most $M_w h_k^3/16$ in absolute value. The fine-grid remainder has bound $C_w(H/2)(\sum h_k^2/2)$, one quarter of the coarse bound. This proves the constant in (S18). Finally, couple the two numerical laws and use $\mathrm{Lip}(\phi_{t_k}) \le C_F$,

[2] AI assistance for this section is disclosed in the acknowledgment [1–3].

followed by the triangle inequality through p_{t_k}. The prescribed law need not be the exact law of the field. Comparing one step with two half steps is the classical step-doubling construction. Here it removes the candidate model's unknown bias and gives a bound that is uniform over the common regularity class. ∎

S3. Proof of the two-grid characterization

The proof first shows that sufficiently accurate coarse and fine runs must approach an exact generator whose acceleration preserves the distribution. It then shows how such a generator permits second-order distribution accuracy. We keep the compact closure because a convergent sequence of smooth fields need not have a smooth limit. The proof permits unequal deterministic step sizes.[3]

A. Necessity: a successful sequence produces an order-elevating generator

First we identify a limiting exact generator. We then show that a test with a nonzero average acceleration effect would contradict the assumed accuracy of the two runs. The contradiction uses a time interval on which that effect has one sign. This is why accuracy is required along the path, rather than only at its endpoint.

Suppose there exist $u_n \in \mathcal{B}_K$, grids with $N_n \to \infty$ and $H_n \to 0$, and $N_n(E_c + E_f) \to 0$. Extract $u_n \to u$ in C^1_{loc} using Lemma S1. By (S10), $\mathfrak{b}(u_n) \to 0$; continuity of the exact laws shows that u generates p on the whole interval.

Assume the acceleration condition fails at some $s \in [0,T]$. Then there is a compact smooth ψ for which $\mathbb{E}[\nabla\psi(X_s)\cdot a_s(X_s)] \neq 0$. Set $\phi = \psi \circ F^u_{0,s}$ and rescale by a positive constant so that its gradient and gradient-Lipschitz bounds are at most one. The flow is $C^{1,1}$ in space, so this is an admissible compactly supported $C^{1,1}$ test. Its transported version equals a positive multiple of ψ at s. For each u_n, transport this *same source test* with that field's own flow. Lemma S1 and the representation $I_{u_n}(t) = \mathbb{E}[\nabla\phi(G)J^{-1}_{n,t}a^{u_n}_t(X^{u_n}_t)]$ show that $I_{u_n} \to I_u$ uniformly in t. Local convergence is uniform on bounded Gaussian initial states; the tail is bounded by a common multiple of $1 + |G|$.

Thus there is a closed interval of positive length ℓ, possibly one-sided at an endpoint, on which all sufficiently large n have I_{u_n} of one sign and $|I_{u_n}| \geq \eta > 0$. Choose the first and last coarse nodes inside this interval. When H_n is small their separation is at least $\ell/2$. Equation (S18) and $C_{\times,w}H_n \leq \eta/8$ yield

$$|Z_j - Z_i| \geq \frac{\eta}{8}\sum_{k=i}^{j-1} h_k^2 \geq \frac{\eta\ell^2}{32N_n}.$$

The last inequality follows from Cauchy–Schwarz. It needs neither a positive lower bound on individual step sizes nor a limit for their relative sizes. But the lemma also gives $|Z_j - Z_i| \leq 2C_F(E_c + E_f) = o(N_n^{-1})$, a contradiction. Therefore $\int \nabla\psi \cdot a_t \, dp_t = 0$ for every compact smooth ψ and every t. This proves necessity for arbitrary refining grids, not only uniform or quasi-uniform ones.

B. Sufficiency: locally Lipschitz acceleration is enough here

Euler omits the leading displacement due to acceleration. To compare distributions, we can pair an exact particle with a different numerical particle, provided each population still has the same distribution. The acceleration flow gives such a pairing. With physical time fixed, it supplies the leading displacement Euler misses while leaving the input distribution unchanged. This change of pairing is a recoupling.

Suppose $u \in \overline{\mathcal{B}}_K$ generates p and satisfies the acceleration condition. Fix t, and let Ψ^r_t be the autonomous flow of a_t. The acceleration is locally Lipschitz with linear growth, so this flow is complete and locally bi-Lipschitz. It preserves p_t. For clarity, p_t is the positive continuous density obtained by pushing Gaussian density through a C^1 bi-Lipschitz flow. The weak identity $\mathrm{div}(p_t a_t) = 0$ extends from smooth tests to compactly supported Lipschitz tests by local mollification. Applied to $\psi \circ \Psi^r_t$, the characteristic chain rule gives $\frac{d}{dr}\int \psi(\Psi^r_t(x))p_t(dx) = 0$ almost everywhere in r. Preimages of compact sets remain compact on bounded r intervals by linear growth. The derivative and the local mollification limits are therefore dominated. Integrating in r proves invariance. Equivalently, this is uniqueness of the stationary continuity equation under a locally Lipschitz complete vector field.

For $X \sim p_t$ use the recoupling $Z = \Psi^{h^2/2}_t(X)$. The new starting point has the same distribution, $Z \sim p_t$. Euler starting at Z therefore has the required one-step output law, but it is now paired differently with the exact particle. Let $G_A = e^{AT^2/2}$. The acceleration estimates imply

$$|Z - X| \leq (AG_A/2)h^2(1 + |X|),$$
$$|Z - X - \tfrac{1}{2}h^2 a_t(X)| \leq (C_D A G_A^2/8)h^4(1 + |X|)^2.$$

[3] AI assistance for this section is disclosed in the acknowledgment [1–3].

The second inequality follows by integrating $|a_t(\Psi_t^r X) - a_t(X)|$ and using the local bound on Da_t along the orbit. Exact-flow Taylor expansion gives

$$F_{t,t+h}(X) = X + hu_t(X) + \tfrac{1}{2}h^2 a_t(X) + R, \quad |R| \le JC_F^2 h^3(1+|X|)^2/6.$$

This remains valid in the closure by passage from smooth fields with the uniform remainder. Consequently

$$|Z + hu_t(Z) - F_{t,t+h}(X)| \le C_{\mathrm{loc}} h^3 (1+|X|)^2, \quad C_{\mathrm{loc}} = C_D A G_A^2 T/8 + KAG_A/2 + JC_F^2/6. \tag{S19}$$

We can now average the particle bound using the Gaussian fourth moment. This gives local W_2 error at most $C_{\mathrm{loc}} C_F^2 \sqrt{M_4} h^3$. Propagating it with Euler's $(1+Kh)$ Lipschitz stability proves, on every grid,

$$\max_k W_2(\widehat{\rho}_k^u, p_{t_k}) \le C_F^3 C_{\mathrm{loc}} \sqrt{M_4} \sum_k h_k^3 \le C_F^3 C_{\mathrm{loc}} \sqrt{M_4}\, TH^2. \tag{S20}$$

The acceleration need not be globally Lipschitz: its derivative grows at most linearly, and the required fourth moment comes from Gaussian initialization and the common linear-growth bound on the velocity.

If u belongs only to the closure, approximate it by a smooth $u_N \in \mathcal{B}_K$ so closely that both finite-grid Euler laws have additional W_2 discrepancy at most N^{-2} at their finitely many common nodes. Such approximation follows from Lemma S1; it need not preserve the target path exactly. On the uniform grid and its bisection, (S20) proves $\mathcal{J}_N(u_N) = O(N^{-2})$. The same approximant is used for the two runs. This proves sufficiency without asserting that a C^1 limit is itself a smooth member of the class.

C. *Uniform lower bound and the three optimized regimes*

The same necessity argument gives a lower bound for all admissible fields on sufficiently fine grids. If no such common bound existed, we could choose a sequence with progressively smaller error. That sequence would produce exactly the generator that is assumed not to exist.

If there were no constants $c, H_0 > 0$ in (4), for each integer m we could select a field and a grid with $H_m \le 1/m$ and $\mathcal{J}_{N_m} < 1/(mN_m)$. Necessarily $N_m \ge T/H_m \to \infty$. The necessity argument would produce an order-elevating generator in the closure, a contradiction. Thus the lower bound is uniform over all fields, all sufficiently fine deterministic grids, and all numbers of steps, not just over a single refinement sequence.

For uniform grids define $q_N = \inf_{u \in \mathcal{B}_K} \mathcal{J}_N(u)$. For each fixed N this infimum equals a minimum over the compact closure, since Euler laws are continuous there. Put $b_* = \min_{u \in \overline{\mathcal{B}}_K} \mathfrak{b}(u)$. Uniformly over that closure,

$$|E_c - \mathfrak{b}(u)| + |E_f - \mathfrak{b}(u)| \le Ch + 2\omega_p(h), \qquad h = T/N,$$

for a class-dependent constant C. This follows from strong Euler error and the difference between a continuous-time supremum and the maximum over common coarse nodes. Therefore $q_N \to 2b_*$. If no exact generator exists, compactness gives $b_* > 0$. If an exact generator exists but no order-elevating generator does, the uniform lower bound gives $q_N \ge c/N$, and approximation of an exact generator, together with strong Euler error, gives $q_N \le C/N$. If an order-elevating generator exists, the previous subsection gives $q_N = O(N^{-2})$. This is a statement about the optimized two-grid error; individual nonoptimal sequences can have other rates.

S4. Gaussian-source topology and the odd-dimensional application

We now specify a path for which the generator required by Theorem 1 cannot exist in odd dimension. Only the first two time derivatives of the path at its Gaussian starting distribution are needed. This pair of derivatives is called the source two-jet. They constrain every exact generator, including nonlinear ones. We can therefore prove nonexistence without first listing all possible generators. The target need not be Gaussian, radial, or unimodal.[4]

Lemma S4 (Source two-jet identity)**.** *Suppose a $u \in \overline{\mathcal{B}}_K$ generates $p_t = \mathrm{Law}((1-t)G + tY)$ near zero, where Y is centered, independent of G, and has covariance S. Let $r(x) = u(0,x) + x$. Then $\mathrm{div}(\gamma_d r) = 0$. For a unit vector e, define $Q_e(y) = \mathbb{E}[(e \cdot r(G))^2 \mid e \cdot G = y]$ and let φ be the standard one-dimensional Gaussian density. For a scalar test ψ with bounded first and second derivatives and at most linear growth,*

$$\mathbb{E}[\psi'(e\cdot G) e \cdot a_0(G)] = \int \psi''(y)\{e^\top S e - Q_e(y)\}\varphi(y)\, dy. \tag{S21}$$

The equality also holds for $C^{1,1}$ tests with almost-everywhere second derivatives, by approximation.

Proof: An exact generator and the prescribed interpolation give the same average of each test wherever their distributions agree. Their first and second time derivatives must therefore agree at zero. For the first derivative, a compact smooth test along the prescribed straight interpolation gives $-\mathbb{E}\nabla\Phi(G)\cdot G$. The derivative along the flow gives $\mathbb{E}\nabla\Phi(G)\cdot u_0(G)$. Equality

[4] AI assistance for this section is disclosed in the acknowledgment [1–3].

says that the correction leaves the Gaussian source unchanged, which is Gaussian solenoidality. Testing along e shows that the distributional derivative of $\varphi(y)\mathbb{E}[e\cdot r(G)\mid e\cdot G=y]$ vanishes. This function is integrable, hence the constant is zero. Lipschitz growth of r and Gaussian conditional expectation give continuous versions of the conditional first and second moments.

For the second derivative, the value along the straight interpolation is $\mathbb{E}\psi''(e\cdot G)(e\cdot(Y-G))^2$, whose conditional squared factor is $y^2+e^\top Se$. Along the flow the second derivative is

$$\mathbb{E}\big[\psi''(e\cdot G)(e\cdot u_0(G))^2+\psi'(e\cdot G)e\cdot a_0(G)\big].$$

The first conditional squared factor equals $y^2+Q_e(y)$ because the conditional mean of $e\cdot r$ is zero. Subtracting the two second-derivative expressions proves the identity. A C^1 field is enough to differentiate the trajectory twice. Linear growth gives the integrable bounds needed to differentiate its expectation. Only the second moment of Y is used. For noncompact scalar tests, first apply spatial cutoffs and then remove them using Gaussian integrability of $|r|^2$, linear growth of a_0, and the bounded test derivatives. Smooth approximation gives the $C^{1,1}$ case without boundary atoms, since the scalar Gaussian has a density. ∎

Lemma S5 (Uniform positive strip test function in odd dimension)**.** *Assume d is odd and $\lambda=\lambda_{\min}(S)>0$. Define*

$$A_d=d+\sqrt{d^2+d-2},\qquad R_K=\max\{1,2A_d(K+1)/\sqrt{\lambda}\},\qquad \delta_K=\frac{\lambda}{2\sqrt{2\pi}}e^{-(R_K+1)^2/2}. \tag{S22}$$

Every exact generator $u\in\overline{\mathcal{B}}_K$ has a unit direction e and a convex strip test $\phi_e(x)=\psi(e\cdot x)$, with gradient and gradient-Lipschitz bounds one, such that $\mathbb{E}[\nabla\phi_e(G)\cdot a_0(G)]\ge\delta_K>0$.

Proof: The identity compares target variance with the mean squared correction in a chosen direction. We seek a region where the latter is too small to cancel the former. A topological argument selects a point on a large sphere. Lipschitz continuity controls the correction near that point. A convex test then detects the acceleration that remains.

Put $L_r=\mathrm{Lip}(r)\le K+1$. The continuous weighted-divergence identity is pointwise and gives $\operatorname{div} r(x)=x\cdot r(x)$, so $|x\cdot r(x)|\le dL_r$. Project the correction onto directions tangent to the sphere of radius R_K. This gives a continuous tangent vector field on a sphere of even dimension $d-1$. The hairy-ball theorem says that this field must vanish somewhere [4]. In $d=1$ the tangent space is zero-dimensional, so the step is immediate. Thus $r(R_Ke)$ is parallel to e for some unit e. We have not assumed that r is tangent elsewhere.

For $y\in[R_K,R_K+1]$ write $G=ye+Z_\perp$ under the scalar conditioning, where $Z_\perp\sim\mathcal{N}(0,I_{d-1})$. Since $Z_\perp\cdot r(R_Ke)=0$, the divergence identity and Lipschitz bound give

$$|e\cdot r(ye+Z_\perp)|\le\frac{L_r}{R_K}\left[d+|Z_\perp|\sqrt{(y-R_K)^2+|Z_\perp|^2}\right].$$

Use $\mathbb{E}|Z_\perp|^2=d-1$ and $\mathbb{E}|Z_\perp|^4=(d-1)(d+1)$ in Minkowski's inequality to obtain $\sqrt{Q_e(y)}\le A_dL_r/R_K\le\sqrt{\lambda}/2$ throughout the strip. Choose ψ' equal to zero below R_K, increasing with slope one on the strip, and equal to one above it. Equation (S21) then gives

$$\mathbb{E}\nabla\phi_e(G)\cdot a_0(G)\ge\frac{3\lambda}{4}\Pr\{R_K\le N(0,1)\le R_K+1\}\ge\frac{3\lambda}{4\sqrt{2\pi}}e^{-(R_K+1)^2/2}\ge\delta_K.$$

The test average is positive, but its lower bound includes a Gaussian tail probability. It can therefore be extremely small for large derivative bounds. The argument establishes positivity, not a useful numerical forecast of the error. ∎

Proof of Theorem 3: An order-elevating generator would give zero average acceleration effect for every compact smooth gradient test. Cutting off the strip test on larger and larger sets and controlling the Gaussian tail gives the same zero value for the strip test in Lemma S5. This contradicts its positive average. The general uniform lower bound in Theorem 1 therefore applies to every field, including fields whose exact law differs from the prescribed interpolation.

One constant in this lower-bound argument remains nonconstructive: we prove that it exists without obtaining a formula for it. Compactness yields $\eta_K>0$ such that $\mathfrak{b}(u)\le\eta_K$ implies a strip test function with source pairing at least $\delta_K/2$. Otherwise a sequence of biases converging to zero would have a C^1_{loc} limit generating p; choose that limit's direction and pass its acceleration pairing to the sequence to obtain a contradiction. Fix

$$\tau=\min\{T,\delta_K/(4M_w)\},\quad H_0=\min\{\eta_K/(2C_{\mathrm{path}}),\tau/2,\delta_K/(32C_{\times,w})\}.$$

If $E_c\ge\eta_K/2$, the lower bound is immediate with $c\le\eta_K/2$. Otherwise (S11) gives $\mathfrak{b}(u)\le\eta_K$. The transported test function has $I_u(t)\ge\delta_K/4$ on $[0,\tau]$. Let t_j be the last node before τ. The prefix version of (S18), with $Z_0=0$, gives

$$C_F(E_c+E_f)\ge|Z_j|\ge\frac{\delta_K}{32}\sum_{k<j}h_k^2\ge\frac{\delta_K\tau^2}{128N}.$$

Thus an admissible c is $\min\{\eta_K/2,\delta_K\tau^2/(128C_F)\}$. The strip location R_K and lower bound δ_K have explicit formulas, but η_K does not. The proof therefore does not give a computable threshold for the general odd-dimensional result. ∎

A. A non-Gaussian target with an admissible exact generator

The lower bound is not only about paths that no regular field can generate. The following mixture has a regular exact generator. In odd dimension, the obstruction still prevents its acceleration from satisfying the required distribution-preserving condition.

For $Y = \mu\xi e_1 + \sigma Z$, with independent symmetric sign ξ, standard d-dimensional Gaussian Z, and $\sigma > 0$, let $q(t) = (1-t)^2 + \sigma^2 t^2$. The canonical conditional velocity is

$$v_t(x) = \frac{(1+\sigma^2)t - 1}{q(t)} x + \frac{\mu(1-t)}{q(t)} \tanh\left(\frac{\mu t x_1}{q(t)}\right) e_1. \tag{S23}$$

This velocity generates the independent path: it averages $Y - G$ conditional on the interpolated position. The function q stays strictly positive on a neighborhood of $[0,1]$, so all scalar coefficient derivatives needed here are bounded. Write the nonlinear term $c(t)\tanh(b(t)x_1)$. Its time derivative grows at most linearly in x; its second time derivative at most quadratically. Spatial derivatives through order two are uniformly bounded. For the mixed derivative, terms such as $b(t)x_1 \tanh''(b(t)x_1)$ are uniformly bounded, since $z\tanh'' z$ is bounded on the real line. All remaining terms are bounded scalar coefficients times bounded hyperbolic derivatives. Hence this field belongs to some finite $\mathcal{B}_K$, including for separated multimodal targets. Thus the class of exact generators can be nonempty for a non-Gaussian target. Positive covariance alone still does not make every $\mathcal{P}_2$ target reachable by a globally Lipschitz flow.

S5. An explicit initial-risk bound without exact marginal preservation

Throughout this section the target path is (9), but u is any field in $\overline{\mathcal{B}}_K$. Put $\nu = \operatorname{tr} S$ and $\Delta_0(u) = \mathbb{E}|u(0,G) + G|^2$. Independence gives the elementary risk identity

$$\mathcal{L}_0(u) = \mathbb{E}|u(0,G) - (Y-G)|^2 = \nu + \Delta_0(u). \tag{S24}$$

Here population loss means an expectation over the source and target distributions, rather than an average over a finite training batch. The Bayes velocity minimizes this squared prediction loss. At the source it is $-x$, with minimum loss ν. The excess risk Δ_0 measures how much loss the candidate adds above that minimum. The decomposition is standard. We use it to constrain a candidate through its two numerical runs, without assuming that its continuous flow preserves the target path.[5]

A. Quadratic transported tests and all constants

We now test the mean squared position, which is the second moment. This test is unbounded and its gradient grows linearly. We therefore keep the moment factors explicitly when repeating the earlier comparison of test averages. The constants below make the final source-risk inequality computable.

Use (S3) and set

$$H_q = (1+KT)C_F^3, \qquad D_q = 3C_F^3\sqrt{M_2}, \qquad M_q = BC_F^4\sqrt{M_2 M_4}, \tag{S25}$$

$$D_0 = AC_F^2 M_2, \qquad c_\ell = \min\{T/D_0, 1/(2M_q)\}, \tag{S26}$$

$$C_{q0} = JC_F^5/6 + KAC_F^3/2 + TH_qA^2C_F^3/4, \tag{S27}$$

$$C_{qb} = H_qA + C_F^2C_D, \qquad C_q = M_3\{C_{q0}C_F^3 + \tfrac{1}{2}C_{qb}C_F^2C_ET\}, \tag{S28}$$

$$C_{\times,q} = \tfrac{5}{4}C_q + M_q/16, \qquad C_H = \max\{2/c_\ell, 16C_{\times,q}\}, \tag{S29}$$

$$C_m = C_F\sqrt{M_2} + \sqrt{d} + \sqrt{\nu}, \qquad J_m = 6KAC_F^2M_2 + 2JC_F^3M_3. \tag{S30}$$

Let $\psi_t(x) = |F_{t,0}(x)|^2/2$ and $I(t) = \mathbb{E}[\nabla\psi_t(X_t)\cdot a_t(X_t)]$. Then

$$|\nabla\psi_t(x)| \le C_F^2(1+|x|), \qquad \|D^2\psi_t(x)\| \le H_q(1+|x|), \tag{S31}$$

$$|I'(t)| \le M_q, \qquad |I(0)| \le D_0. \tag{S32}$$

The derivative bound follows from (S7). Take the dot product of transported acceleration with G, then use Cauchy–Schwarz and the Gaussian moment bounds. The same constant also gives a conservative bound when the test is transported to any other fixed reference time.

The proof of Lemma S2, now with a linearly growing Hessian, gives a one-step remainder at most $C_{q0}h^3(1+|x|)^3$. To see the three terms, the trajectory Taylor remainder costs $JC_F^5h^3(1+|x|)^3/6$, the transport of the gradient costs at most $KAC_F^3h^3(1+|x|)^2/2$, and the quadratic Taylor remainder costs at most $TH_qA^2C_F^3h^3(1+|x|)^3/4$. The scalar measurement $\nabla\psi_t\cdot a_t$ has derivative bounded by $C_{qb}(1+|x|)^2$. At numerical and exact states its expectation difference is at most $C_{qb}C_F^2C_EM_3S_{2,k}$. Summation and $S_{2,k} \le TH$ yield the interval weak remainder $C_qH\sum h_k^2$. Consequently the two-grid contrast formula (S18) holds with $C_{\times,q}$ in place of $C_{\times,w}$.

[5] AI assistance for this section is disclosed in the acknowledgment [1–3].

Although ψ_t is not globally Lipschitz, for the two numerical laws α, β at the same time it satisfies

$$\left| \int \psi_t \, d\alpha - \int \psi_t \, d\beta \right| \le D_q W_2(\alpha, \beta) \le D_q(E_c + E_f). \tag{S33}$$

Indeed the mean-value inequality bounds the integrand by $C_F^2(1 + |x| + |y|)|x - y|$; both numerical states have root second moment at most $C_F\sqrt{M_2}$. Cauchy–Schwarz gives the conservative constant D_q. The target is used only in the Wasserstein triangle inequality, so no fourth moment of Y is required here.

Put $a_* = |I(0)| = |\mathbb{E}G \cdot a_0(G)|$. If $a_* > 0$, on $[0, \ell]$ with $\ell = \min\{T, a_*/(2M_q)\} \ge c_\ell a_*$ the pairing has a common sign and magnitude at least $a_*/2$. If $H \le \ell/2$ and $H \le a_*/(16C_{\times,q})$, choose the last node before ℓ. The contrast and (S33) give

$$D_q(E_c + E_f) \ge \frac{a_*}{16} \sum_{k<j} h_k^2.$$

Cauchy–Schwarz gives

$$a_* \le \left\{ \frac{64 D_q N (E_c + E_f)}{c_\ell^2} \right\}^{1/3}. \tag{S34}$$

On the uniform grid, $\sum_{k<j} h_k^2 = (T/N) t_j \ge T\ell/(2N)$, giving

$$a_* \le \left\{ \frac{32 D_q N (E_c + E_f)}{T c_\ell} \right\}^{1/2}. \tag{S35}$$

If either small-mesh condition fails, then $a_* \le C_H H$ instead. Adding this term makes each displayed bound valid without a mesh qualification. If $a_* = 0$, the same statements hold trivially.

B. Recovering the first two source-time moment derivatives

To use the source-risk identity, we need the first two time derivatives of the model's second moment at time zero. Accuracy along the path controls the moment itself. A common bound on its third derivative then controls these two source-time derivatives. We obtain explicit bounds by using forward finite differences, which estimate derivatives from nearby times.

Let $m_u(t) = \mathbb{E}|X_t^u|^2$, $m_p(t) = d(1-t)^2 + \nu t^2$, and $f = m_u - m_p$. Then $f(0) = 0$ and

$$\|f\|_\infty \le C_m \mathfrak{b}(u), \qquad |f'''(t)| \le J_m \quad \text{almost everywhere.} \tag{S36}$$

The square roots of second moments differ by at most the Wasserstein distance. This inequality gives the first bound. The second follows from $m_u''' = 6\mathbb{E}[u_t \cdot a_t] + 2\mathbb{E}[X_t \cdot j_t]$ and (S6). For a closure field the inequality follows by passing the uniform finite-difference estimates of smooth approximants; classical continuous third derivatives are not being assumed.

For $0 < s \le T/2$, forward differences and the Lipschitz bound on f'' give

$$|f'(0)| \le \frac{5 C_m \mathfrak{b}(u)}{2s} + J_m s^2, \tag{S37}$$

$$|f''(0)| \le \frac{3 C_m \mathfrak{b}(u)}{s^2} + J_m s. \tag{S38}$$

The first formula uses $(4f(s) - f(2s))/(2s)$; the second uses $(f(2s) - 2f(s) + f(0))/s^2$. For the latter, the double-integral kernel has average evaluation time s, so a J_m-Lipschitz second derivative incurs at most $J_m s$. Since $s \le 1$,

$$|f'(0) + f''(0)/2| \le \omega_K(\mathfrak{b}(u)), \quad \omega_K(b) = \inf_{0<s\le T/2} \left\{ \frac{4 C_m b}{s^2} + \frac{3}{2} J_m s \right\}. \tag{S39}$$

The infimum chooses the finite-difference spacing that gives the smallest bound. Here $\omega_K(0) = 0$, and for $b > 0$ the minimizer is $s = \min\{T/2, (16 C_m b/(3J_m))^{1/3}\}$. This explicit modulus is nondecreasing and $O(b^{1/3})$ as $b \downarrow 0$.

The exact source identity, valid for every field, is

$$\mathbb{E}[G \cdot a_0(G)] = \nu - \Delta_0(u) + f'(0) + \tfrac{1}{2} f''(0). \tag{S40}$$

To verify it, note that $m_u'(0) = 2\mathbb{E}[G \cdot u_0(G)]$ and $\mathbb{E}|u_0(G)|^2 = d - f'(0) + \Delta_0(u)$. Insert these into $m_u''(0) = 2\mathbb{E}|u_0(G)|^2 + 2\mathbb{E}[G \cdot a_0(G)]$ and use $m_p''/2 = d + \nu$. This calculation does not require Gaussian solenoidality or exact path preservation.

Proof of Theorem 4: Combine (S40), (S39), (S11), and either (S34) or (S35), including the $C_H H$ alternative. This gives precisely Eqs. (12)–(13). If $H_N \to 0$ and $E_c + E_f = o(N^{-1})$, every term tends to zero, so $\Delta_0(u_N) \to \nu$ and $\mathcal{L}_0(u_N) \to 2\nu$. For a fixed gap $|\Delta_0 - \nu| \ge g > 0$, an entirely explicit choice is

$$s_g = \min\{T/2, g/(12J_m)\}, \qquad b_g = g s_g^2/(32 C_m), \tag{S41}$$

$$H_g = \min\{T, g/(4C_H), b_g/(2C_{\mathrm{path}})\}, \qquad c_g = \min\{b_g/2, c_\ell^2 g^3/(4096 D_q)\}. \tag{S42}$$

If $H \le H_g$ and $\mathcal{J}_N < c_g/N$, then $E_c + C_{\mathrm{path}} H \le b_g$. Evaluating the infimum in ω_K at s_g gives at most $g/4$, the mesh term is at most $g/4$, and the cubic term is at most $g/4$. The asserted gap would therefore be at most $3g/4$, a contradiction. Thus the risk-gap barrier is uniform over all sufficiently fine deterministic grids, with no lower step bound or condition $H = O(N^{-1})$. Although every constant has a formula, the bounds may be far too conservative to give a useful numerical certificate. ∎

S6. EXACT EXAMPLES, LOSS OF A FIT AFTER REFINEMENT, AND NECESSARY RESTRICTIONS

A. An exact fit on one Euler grid can fail after refinement

The next example separates two questions: does Euler follow the fitted velocity accurately, and does its output match the target? We choose a velocity for which Euler matches the target exactly at one step size. The exact continuous flow of that velocity does not match the target. Using smaller steps therefore exposes the model error that the coarse numerical error had canceled.[6]

Let $g(t) = \sqrt{(1-t)^2 + \sigma^2 t^2}$, $\sigma > 0$, and $u_h(t,x) = c_h(t)x$, where $c_h(t) = (g(t+h) - g(t))/(hg(t))$. On a fixed neighborhood of $[0,1]$, g is positive, so this family has a common derivative bound as $h \downarrow 0$. The exact algebraic identity

$$1 + hc_h(t) = g(t+h)/g(t)$$

shows why the coarse computation is exact at every target node. The scale factors multiply, and each denominator cancels the preceding numerator. This gives every target node marginal exactly in every dimension. The exact scale of the continuous model is $q_h(t) = \exp\{\int_0^t c_h(s)ds\}$ and Taylor expansion gives

$$q_h(t) = g(t) \exp\left\{ \frac{h}{2} \int_0^t \frac{g''(s)}{g(s)} ds + O(h^2) \right\}. \tag{S43}$$

Now keep this *same* u_h and use Euler step $h/2$. For a smooth scalar coefficient c, the logarithm of the Euler scale on step k is

$$\int_0^t c(s)ds - \frac{k}{2} \int_0^t \{c'(s) + c(s)^2\} ds + O(k^2).$$

To obtain this expression, approximate the integral by its left Riemann sum and expand $\log(1+kc)$. The coefficient family and the derivatives used in this calculation have common bounds. The remainder bounds are therefore uniform in h. With $c_h = g'/g + (h/2)g''/g + O(h^2)$ and $k = h/2$, the refined scale satisfies

$$\widehat{g}_{h/2;u_h}(t) = g(t) \exp\left\{ \frac{h}{4} \int_0^t \frac{g''(s)}{g(s)} ds + O(h^2) \right\} \tag{S44}$$

at common nodes. Since $g''/g = \sigma^2/g^4 > 0$, the refined target error is $\Theta(h)$ even though the fitted coarse error is exactly zero. The same computation for step h/m has coefficient $(h/2)(1-1/m)$, for any fixed integer $m \geq 2$. Thus a velocity chosen for one grid can attain the first-order lower bound. This calculation explains the cancellation, but it does not give a general learning algorithm. If a network receives the requested step size as an input, changing that input can change the evaluated field. Such a comparison falls outside the same-field requirement here.

For the convex blend $c_{h,\theta} = (1-\theta)g'/g + \theta c_h$, the leading logarithmic target errors on steps h and $h/2$ have coefficients $(\theta - 1)h/2$ and $(\theta - 1/2)h/2$ respectively. Their largest absolute coefficient is minimized at $\theta = 3/4$, with strictly positive value $h/8$. Fitting both resolutions within this scalar family therefore reduces the error constant while leaving the order unchanged. The numerical one-dimensional bounded optimization checks this fact for the stated family, not for all neural fields.

B. Even-dimensional exactness and insufficiency of scalar risk

The restriction to odd dimension matters. In even dimension, a rotation can change which source and target points are paired while preserving the Gaussian distribution at every time. The resulting trajectories can be straight, so Euler can be exact. The following construction provides this contrasting case.

For even d choose $J^\top = -J$, $J^2 = -I$, and set $A_t = (1-t)I + \sigma t J$. The process $A_t G$ has covariance $g(t)^2 I$, with straight trajectories. Its velocity $u_t = (\sigma J - I)A_t^{-1}x$ has zero pointwise acceleration, belongs to a class $\mathcal{B}_K$ with finite K, and Euler is exact on every grid. This skew-coupling construction is known and is credited to the existing Gaussian geometry rather than counted as a new theorem [5, 6]. At the source, $u_0 + \mathrm{id} = \sigma J x$, so $\Delta_0 = d\sigma^2 = \nu$ and the doubled-risk necessity is attained.

Matching the required source loss alone is not sufficient to make acceleration distribution-preserving. In $d = 3$, take $S = I_3$, $g^2 = (1-t)^2 + t^2$, and

$$u_t(x) = \left\{ (g'/g)I + \sqrt{3/2}\, g^{-2} J_{12} \right\} x,$$

where J_{12} is the unit rotation on coordinates $1, 2$ and is zero on coordinate 3. This is an exact generator. Its source excess is $3 = \nu$, but $a_0(x) = \mathrm{diag}(-1/2, -1/2, 1)x$. The quadratic test $x_3^2/2$ detects a positive average acceleration effect, so the acceleration condition fails. Thus neither the loss equality nor small target errors at one or two finite step counts establish the existence of an order-elevating generator.

[6] AI assistance for this section is disclosed in the acknowledgment [1–3].

C. Why endpoint-only tests are excluded

An acceleration contribution can be positive during one part of the interval and negative during another. These contributions can cancel at the final time while errors remain at intermediate times. The following scalar example shows why testing only the final distribution cannot replace testing the whole path.

An explicit scalar example is $X_t = f(t)G$, with $u(t,x) = (f'(t)/f(t))x$ and $f > 0$. The leading terminal Euler coefficient is proportional to $\int_0^T f''(s)/f(s)\,ds$; it may cancel while the prefix integrals do not. For example, on $[0,1]$ take $f(t) = \exp\{-5t^3/3\}$. Then $u(t,x) = -5t^2x$ and $\int_0^1 f''/f = -5 + 25/5 = 0$, although f''/f is not identically zero. Thus terminal Euler error is second-order for this example while whole-path error is first-order. The necessity proof compares two times surrounding an interval on which the average acceleration effect is nonzero and keeps one sign. Final-time data alone cannot provide those two-time comparisons. Existing finite-step Gaussian terminal-cancellation constructions likewise leave intermediate marginals unconstrained [6]. None of the main results rules out accurate endpoint-only generators, other numerical methods, state augmentation, or step-conditioned average-velocity models.

S7. Finite-step observability and nondegenerate refinements

The candidate has its own exact distribution at each time, which may differ from the target. We use two numerical runs to bound how much acceleration can change that exact distribution. Lemma S2 controls one test average at a time. To control the full component of acceleration that changes the distribution, we must also bound the part that the chosen tests leave unmeasured.[7]

A. How the interval splits determine the comparison

Adding more time points need not substantially change a numerical run. For example, a split can leave one subinterval almost as long as the coarse interval. The refinement energy below measures the reduction in squared step sizes. A positive lower bound makes the two runs different enough to reveal the leading error.

Split the kth coarse interval into positive fractions $\theta_{k,1}, \dots, \theta_{k,m_k}$ summing to one. Define

$$\beta_k = 1 - \sum_r \theta_{k,r}^2, \qquad \beta_k \ge \beta_0 > 0. \tag{S45}$$

The error E_r is measured at common coarse nodes; set $E = E_c + E_r$. For any transported test with acceleration pairing $I(t)$, Lipschitz constant M, and weak-remainder constant C_w, direct subtraction gives

$$Z_j - Z_i = -\frac{1}{2}\sum_{k=i}^{j-1} \beta_k h_k^2 I(t_k) + R_{ij}, \qquad |R_{ij}| \le C_\times H \sum_{k=i}^{j-1} h_k^2, \quad C_\times = 2C_w + M/2. \tag{S46}$$

Indeed the fine leading sum is $-\frac{1}{2}\sum_{k,r} \theta_{k,r}^2 h_k^2 I(t_k + s_{k,r})$, with $0 \le s_{k,r} \le h_k$. Replacing each pairing by $I(t_k)$ costs at most $(M/2)\sum h_k^3$. The two recursive remainders together cost at most $2C_w H \sum h_k^2$. No numerical trajectory is restarted at an exact marginal. For equal m-fold subdivision a smaller constant is available:

$$\beta_0 = 1 - 1/m, \qquad C_\times = (1 + m^{-2})C_w + \frac{m-1}{4m^2}M. \tag{S47}$$

Lemma S6 (Explicit observation of a signed residual)**.** *Suppose* $|I| \le A_o$, $\mathrm{Lip}(I) \le M$, $|Z_k| \le DE$, *and* (S46) *holds. Assume* $A_o, M > 0$ *and put*

$$c_o = \min\{T/(2A_o), 1/(2M)\}, \qquad L_o = \max\{4/c_o, 8C_\times/\beta_0\}.$$

Then, on any coarse grid,

$$\|I\|_\infty \le L_o H + \left(\frac{64DNE}{\beta_0 c_o^2}\right)^{1/3}. \tag{S48}$$

On a uniform coarse grid this improves to

$$\|I\|_\infty \le L_o H + \left(\frac{32DNE}{\beta_0 T c_o}\right)^{1/2}. \tag{S49}$$

Zero values of A_o *or* M *are handled directly or by increasing them to any positive upper bounds.*

Proof: The proof uses the fact that I cannot change arbitrarily quickly. If its magnitude is large at one time, it stays away from zero with the same sign on a nearby interval. The difference between the two runs on that interval then forces a lower bound on their target errors.

[7] AI assistance for this section is disclosed in the acknowledgment [1–3].

Let $a = \|I\|_\infty > 0$ and choose a maximizing time. On its longer side there is an interval of length $\ell = \min\{T/2, a/(2M)\} \geq c_o a$ where I has one sign and $|I| \geq a/2$. If $H \leq \ell/4$, its first and last coarse nodes are separated by at least $\ell/2$. If also $H \leq \beta_0 a/(8C_\times)$, (S46) gives

$$2DE \geq |Z_j - Z_i| \geq \frac{\beta_0 a}{8} \sum_{k=i}^{j-1} h_k^2 \geq \frac{\beta_0 a \ell^2}{32N}.$$

This proves the cubic bound. On a uniform grid the last squared-step sum is at least $T\ell/(2N)$, proving the square-root bound. If either mesh condition fails, $a \leq L_o H$. Adding the two alternatives proves the statements for every grid, including coarse grids where the bound is weak. ∎

For source tests with $\|\nabla\phi\|_\infty, \mathrm{Lip}(\nabla\phi) \leq 1$, the constants in Supplementary Section S1 and Lemma S2 can be used directly. Taking the supremum over all such tests bounds the largest effect detected by this test class at any time. This is a weak acceleration residual. It is not yet an L^2 bound on the whole acceleration component.

B. Bounding the full residual with a Gaussian polynomial expansion

A finite collection of measurements can miss a nonzero acceleration component. We therefore also bound the part that is not measured. We express the transported acceleration as a sum of orthogonal Gaussian polynomial components. A derivative estimate bounds the contribution of the omitted higher-degree components, called the spectral tail. This argument uses standard Gaussian initialization. Write

$$X_t = F_{0,t}(G), \quad J_t = DF_{0,t}, \quad V_t = J_t^{-1} a_t \circ F_{0,t}, \quad \varepsilon_t = \|\Pi_{\rho_t^u} a_t\|_{L^2(\rho_t^u)}.$$

Here J_t is the spatial Jacobian, which describes how the flow changes when its initial position changes. The field V_t expresses acceleration in those initial coordinates. The projection $\Pi_{\rho_t^u}$ keeps the component detected by gradients of test functions. Its norm ε_t measures the part that can change the current distribution. We use the constants $C_F, A, B, C_D, C_E, M_2, M_4$ from (S3)–(S6). Define

$$A_o = AC_F^2\sqrt{M_2}, \qquad M = BC_F^3\sqrt{M_4}, \tag{S50}$$

$$c_o = \min\{T/(2A_o), 1/(2M)\}, \qquad V_1 = KTAC_F^5 + C_D C_F^3. \tag{S51}$$

Let H_α be normalized multivariate probabilists' Hermite polynomials, so $\mathbb{E}H_\alpha(G)H_\eta(G) = \mathbf{1}_{\alpha=\eta}$. These polynomials are orthogonal under the standard Gaussian law, so the mean product of two different basis polynomials is zero. The multi-index α lists the degree in each coordinate, and $|\alpha|$ is their total degree. For $\ell = |\alpha| \geq 1$ set $\phi_\alpha = H_\alpha/\sqrt{\ell}$. In the polynomial envelopes below, $m = \ell$ denotes this total degree; it is distinct from the subdivision factor in (S47). If $\phi_\alpha(x) = \sum_b c_b x^b$, use the explicitly finite envelopes

$$P_{1,\alpha} = \sum_b |c_b||b|, \qquad P_{2,\alpha} = \sum_b |c_b||b|(|b| - 1).$$

Then $|\nabla\phi_\alpha(x)| \leq P_{1,\alpha}(1 + |x|)^{m-1}$ and $\|D^2\phi_\alpha(x)\| \leq P_{2,\alpha}(1 + |x|)^{\max(m-2,0)}$. Set $\mu_r = \mathbb{E}(1 + |G|)^r$, including $\mu_0 = 1$. Every μ_r is explicit:

$$\mu_r = \sum_{j=0}^{r} \binom{r}{j} 2^{j/2} \frac{\Gamma((d+j)/2)}{\Gamma(d/2)}.$$

The next constants keep track of how each polynomial and its derivatives grow after transport by the flow. They make the coefficient bounds explicit. For each α define

$$G_\alpha = P_{1,\alpha} C_F^\ell, \qquad H_\alpha^* = (P_{2,\alpha} + KTP_{1,\alpha}C_F)C_F^\ell, \tag{S52}$$

$$D_\alpha = G_\alpha 2^{m-1} C_F^{m-1}\sqrt{\mu_{2m-2}}, \tag{S53}$$

$$C_{0,\alpha} = JC_F^3 G_\alpha/6 + KAC_F G_\alpha/2 + TH_\alpha^* A^2 C_F^{\ell+1}/8, \tag{S54}$$

$$C_{b,\alpha} = H_\alpha^* A + G_\alpha C_D, \tag{S55}$$

$$C_{w,\alpha} = \mu_{\ell+1}\{C_{0,\alpha}C_F^{\ell+1} + \tfrac{1}{2}C_{b,\alpha}C_F^m C_E T\}. \tag{S56}$$

Use (S46) or (S47) to define $C_{\times,\alpha}$, and put

$$L_\alpha = \max\{4/c_o, 8C_{\times,\alpha}/\beta_0\}, \tag{S57}$$

$$b_\alpha(E, H) = L_\alpha H + \left(\frac{64 D_\alpha NE}{\beta_0 c_o^2}\right)^{1/3}. \tag{S58}$$

On a uniform coarse grid replace the cube-root term by $\{32D_\alpha NE/(\beta_0 T c_o)\}^{1/2}$.

Theorem S7 (Explicit spectral residual observability)**.** *For every $u \in \overline{\mathcal{B}}_K$, every balanced refinement, and every integer $q \geq 0$,*

$$\sup_{0\leq t\leq T} \varepsilon_t \leq C_F \left\{ \sum_{1\leq|\alpha|\leq q+1} b_\alpha(E,H)^2 + \frac{dM_2V_1^2}{q+1} \right\}^{1/2}. \tag{S59}$$

The infimum over q is a fully specified upper bound. The acceleration and projection refer to the field's own exact law; exact target-path preservation is not assumed. The same statement is valid if E is replaced by the maximum cross-grid W_2 distance, which may be smaller than $E_c + E_r$.

Proof: We first apply the signed observer to each polynomial coefficient. We then bound the coefficients beyond the retained degree and carry the resulting norm bound from source coordinates to the current law. This second step is what turns finitely many coefficient estimates into a bound for the full residual.

For $\phi = \phi_\alpha$, let $\phi_t = \phi \circ F_{t,0}$. Its gradient and Hessian are bounded by $G_\alpha(1+|x|)^{m-1}$ and $H^*_\alpha(1+|x|)^{m-1}$. The second bound follows from the two terms in the second derivative of a composition and $\|D^2F_{t,0}\| \leq KTC_F^2$. A Taylor expansion of ϕ_{t+h} at the exact endpoint, comparing it to the Euler endpoint, gives a remainder at most $C_{0,\alpha}h^3(1+|x|)^{\ell+1}$. The three terms in (S56) bound three sources of error. They are the change in particle acceleration, the transport of the test gradient, and the Taylor remainder involving the Hessian. The Hessian contains the test function's second spatial derivatives. For example, along the segment between the two endpoints, $1+|z| \leq C_F(1+|x|)$ and $|E-Y| \leq AC_Fh^2(1+|x|)/2$; its Hessian remainder is therefore at most $H^*_\alpha A^2 C_F^{\ell+1}h^4(1+|x|)^{\ell+1}/8$.

The pairing $\nabla\phi_t \cdot a_t$ has spatial derivative bounded by $C_{b,\alpha}(1+|x|)^\ell$. Comparing recursive numerical states to exact states and then summing over any subinterval gives exactly $C_{w,\alpha}H\sum h_k^2$, using $\mathbb{E}(1+|G|)^{\ell+1} = \mu_{\ell+1}$. Thus all recursive estimates continue to hold for these unbounded polynomial tests; no target moment beyond the Wasserstein comparisons is used. For two numerical laws, the mean-value inequality and Cauchy–Schwarz give

$$|\mathbb{E}\phi_t(\widehat{X}) - \mathbb{E}\phi_t(\widetilde{X})| \leq D_\alpha W_2(\operatorname{Law}\widehat{X}, \operatorname{Law}\widetilde{X}).$$

To check the constant, the squared gradient factor has moment at most $G_\alpha^2 2^{2m-2}C_F^{2m-2}\mu_{2m-2}$ under any coupling, using only the separate Gaussian-start marginal moment bounds.

The Hermite normalization gives $\|\nabla\phi_\alpha\|_{L^2(\gamma_d)} = 1$. The coefficient $c_\alpha(t) = \mathbb{E}\nabla\phi_\alpha(G)\cdot V_t(G)$ therefore satisfies $|c_\alpha| \leq A_o$ and $\operatorname{Lip}(c_\alpha) \leq M$, because $|V_t(G)| \leq AC_F^2(1+|G|)$ and $\partial_t V_t = J_t^{-1}(j_t - Du_t a_t)\circ F_{0,t}$ has L^2 norm at most $BC_F^3\sqrt{M_4}$. Lemma S6 yields $\sup_t|c_\alpha(t)| \leq b_\alpha$.

We must also account for components outside the retained finite set. The gradients $\nabla H_\alpha/\sqrt{|\alpha|}$ form an orthonormal basis of the Gaussian gradient subspace. Thus their coefficients account for the full projected field, not only the measurements kept in the sum. Orthogonality follows from $\partial_i H_\alpha = \sqrt{\alpha_i}H_{\alpha-e_i}$. Completeness follows by expanding a smooth compact potential in Hermite polynomials in the Gaussian Sobolev norm; integration by parts identifies its derivative coefficients. Polynomial gradients themselves belong to the closure of compact smooth gradients by Gaussian-tail cutoffs. Gaussian differentiation also gives $\sum_{i,b}|b|\,|\langle V_i, H_b\rangle|^2 = \mathbb{E}\|DV\|_{\mathrm{F}}^2$ for $V \in W^{1,2}(\gamma_d)$, first for polynomials and then by approximation in that Sobolev norm. Here the Gaussian Sobolev norm controls both the square-integrable function and its square-integrable first derivatives under the Gaussian law. Since projection onto gradients preserves each total Hermite degree,

$$\|\Pi_{\gamma_d}V_t\|_2^2 \leq \sum_{1\leq|\alpha|\leq q+1}|c_\alpha(t)|^2 + \frac{\mathbb{E}\|DV_t\|_{\mathrm{F}}^2}{q+1}.$$

The chain rule, $D(J^{-1}) = -J^{-1}(DJ)J^{-1}$, and the flow estimates give $\|DV_t(x)\|_{\mathrm{op}} \leq (KTAC_F^5 + C_DC_F^3)(1+|x|)$, hence the numerator is at most $dM_2V_1^2$. Finally, for any physical test ψ and $\phi = \psi\circ F_{0,t}$,

$$\langle\nabla\psi, a_t\rangle_{\rho_t^u} = \langle\nabla\phi, V_t\rangle_{\gamma_d}, \qquad \|\nabla\phi\|_2 \leq C_F\|\nabla\psi\|_{L^2(\rho_t^u)}.$$

Taking the dual supremum proves $\varepsilon_t \leq C_F\|\Pi_{\gamma_d}V_t\|_2$ and the theorem. All derivative estimates are valid almost everywhere for closure fields; the weak expansion passes from smooth approximants with the stated uniform remainders. ∎

The cutoff degree q sets how many coefficients enter the finite sum. For fixed q, all constants are finite and explicit. They can nevertheless grow rapidly with q, K, and dimension, so the bound is not a practical dimension-free certificate.

The order of choices explains what the bound does establish. First choose q large enough to make the tail small. Then make H and NE small enough to control the retained coefficients. Consequently, $H \to 0$ and $NE \to 0$ force the *full* projected-acceleration norm to zero, uniformly in time. This conclusion does not assume that a finite dictionary alone detects every component. It also gives no power-law estimate of distance to the set of exact generators.

a) Converting a residual lower bound into an error lower bound.: Suppose a separate argument proves that the projected acceleration cannot be arbitrarily small. The upper bound just obtained then prevents the two-grid error from being arbitrarily small. The construction below gives explicit constants for this implication.

Fix a proved floor $\sup_t \varepsilon_t \geq \delta > 0$. Choose an integer q with $q+1 \geq 4C_F^2 dM_2 V_1^2/\delta^2$. Write $b_\alpha = L_\alpha H + \Lambda_\alpha (NE)^{1/p}$, where $p = 3$ for arbitrary balanced grids and $p = 2$ for uniform grids, and let $L_q = (\sum L_\alpha^2)^{1/2}$ and $\Lambda_q = (\sum \Lambda_\alpha^2)^{1/2}$ over the retained indices. Then

$$H \leq H_\delta := \frac{\delta}{4C_F L_q} \quad \Longrightarrow \quad E \geq \frac{c_\delta}{N}, \qquad c_\delta = \left(\frac{\delta}{4C_F \Lambda_q}\right)^p.$$

Indeed, a smaller error makes the retained coefficient norm at most $\delta/(2C_F)$ and the tail at most $\delta^2/(4C_F^2)$, contradicting the floor via (S59). All constants are explicit. A proved positive acceleration pairing can supply such a floor; a target-feature discrepancy alone cannot. The constants and number of retained coefficients may be impractical.

C. Why bounded tests alone do not give the full norm

A test average can be small because positive and negative contributions cancel. The squared norm does not have this cancellation. A rapidly oscillating acceleration can therefore have very small smooth-test averages while its norm is larger. The example below shows why no common linear bound compares these quantities under the stated derivative bound.

For $d = 1$, $u_n(t,x) = t\sin(nx)/n^2$ belongs to $\mathcal{B}_1$ for $0 \leq t \leq 1$, and $a_0 = \sin(nx)/n^2$. Its full source residual is

$$\|a_0\|_2 = \frac{\sqrt{(1-e^{-2n^2})/2}}{n^2}.$$

In contrast, Gaussian integration by parts gives, for $|\phi'|, |\phi''| \leq 1$,

$$|\mathbb{E}\phi'(G)a_0(G)| \leq (1 + \mathbb{E}|G|)/n^3.$$

The flow of u_n satisfies $F_{0,t}(x) = x + O(n^{-2})$ and $DF_{0,t}(x) = 1 + O(n^{-1})$ uniformly for $0 \leq t \leq 1$, so the transported pairings are $O(n^{-3})$ at every time. Hence no common linear bound of the full norm by this weak residual holds, even within the stated derivative bound. A finite fixed dictionary can also miss an entire nonzero orthogonal component. The spectral tail in (S59), rather than an unproved norm equivalence, repairs that logical gap.

S8. Refinement and Runge–Kutta versions of the characterization

The characterization depends on separating the leading errors of the two numerical runs. Balanced interval subdivision supplies that separation for Euler. A suitable pair of Runge–Kutta methods can supply it on one common grid. We state the conditions for each comparison and then give examples showing why weaker conditions do not suffice.[8]

With (S45), the necessity proof of Theorem 1 applies with the same steps after replacing its contrast by (S46). A nonzero limiting pairing retains one sign on a fixed interval of length ℓ, and $\sum h_k^2 \geq \ell^2/(4N)$ there. The constant β_0 must be uniform in the sequence. Sufficiency is unchanged since the fine maximum step is at most H. Thus all equivalences and the uniform c/N conclusion hold for every fixed integer refinement factor and every uniformly balanced, possibly interval-dependent subdivision. Fine step counts may differ from $2N$; the reference step count is the coarse number of steps.

A. Pairs of Runge–Kutta methods with distinct leading defects

A Runge–Kutta update evaluates the velocity at several intermediate stage states before forming the next state. Its coefficients specify those states and the weighted output. The leading defect is the first local error term relative to the exact update. The scalar coefficient below determines that term under the stated consistency condition.

Consider a fixed explicit Runge–Kutta method with coefficients A_{ij}, b_i, $c_i = \sum_j \mathsf{A}_{ij} \in [0,1]$, and $\sum_i \mathsf{b}_i = 1$. All coefficients and the number of stages are independent of N. Write

$$\kappa = \tfrac{1}{2} - \sum_i \mathsf{b}_i c_i.$$

The internal consistency $c = \mathsf{A}\mathbf{1}$ is essential when the velocity depends explicitly on time. It makes the two second-order terms involving time and space derivatives have the same coefficient. Taylor expansion of the stages gives

$$\Psi_h^u(t,x) = x + hu(t,x) + h^2(\tfrac{1}{2} - \kappa)a_t(x) + O(h^3(1+|x|)^2). \tag{S60}$$

The remainder bound holds for every field in $\mathcal{B}_K$ with the same constants. Applying the estimates stage by stage gives common linear-growth and Lipschitz bounds $1 + Ch$ for $h \leq H_0$. The constants depend only on K, T and the fixed tableau. The C^2 derivative bound bounds each Taylor remainder. Gaussian fourth moments then bound its L^2 norm.

[8] AI assistance for this section is disclosed in the acknowledgment [1–3].

Theorem S8 (Leading-defect separation for a solver pair)**.** *Run two such methods on the same deterministic N-step grid with the same field, and let their constants satisfy $\kappa_1 \neq \kappa_2$. Replace $E_c + E_f$ in Theorem 1 by the sum of these methods' maximum-node target W_2 errors. All three equivalences and the uniform c/N alternative remain valid. The common derivative bound and target path are unchanged; the fields and grids may depend on N.*

Proof: For a transported bounded-gradient test, (S60) and the same recursive state comparison as Lemma S2 give

$$D_j^{(l)} - D_i^{(l)} = -\kappa_l \sum_{k=i}^{j-1} h_k^2 I(t_k) + O\left(H \sum_{k=i}^{j-1} h_k^2\right), \quad l = 1, 2.$$

Subtracting cancels the model law and retains coefficient $\kappa_1 - \kappa_2 \neq 0$. All constants are common to the derivative class. The constants can depend on the full fixed tableaus. The strong first-order bound (S9) and the node-to-path bound (S10) hold for each method with such constants, for $H \leq H_0$. Strong first-order convergence on arbitrary refining grids gives an exact limiting generator whenever the two target errors vanish. A nonzero acceleration pairing in that generator persists on an interval; the signed contrast and $\sum h_k^2 \geq \ell^2/(4N)$ contradict superlinear two-method accuracy, exactly as in Supplementary Section S3.

Conversely, suppose an order-elevating generator belongs to the closure. We change how numerical and exact particles are paired without changing their distributions. For method l, move the exact particles along the acceleration field for auxiliary time $\kappa_l h^2$, while physical time stays fixed. This auxiliary time may be negative. Linear growth makes the acceleration flow complete in both directions. Inserting the recoupled state into (S60) gives the exact trajectory through order two, with local W_2 defect $O(h^3)$; the local Lipschitz and moment estimates are the same as the Euler recoupling proof up to tableau constants. Method stability propagates the error to $O(H^2)$. Smooth approximants to a generator in the closure approximate both finite numerical runs simultaneously. The contradiction proof of a common lower bound is then unchanged. ∎

Euler has $\kappa = 1/2$ and Heun has $\kappa = 0$, so Euler–Heun on the same grid is a concrete instance, using N and $2N$ field evaluations. Two second-order Runge–Kutta methods both have $\kappa = 0$ and do *not* satisfy this theorem: ordinary second-order accuracy already allows superlinear two-grid accuracy for any sufficiently regular exact generator. The theorem is a class of solver-pair converses, not a claim that every integrator has the Euler obstruction.

Distinct leading defects are needed even when both methods have first-order accuracy. Take an explicit three-stage tableau with $c = (0, 1/2, 1)$, $\mathsf{A}_{21} = 1/2$, $\mathsf{A}_{31} = 1$, other entries zero, and $\mathsf{b} = (1 + \eta, -2\eta, \eta)$ for fixed $\eta \neq 0$. It is consistent, has $\mathsf{b}^\top c = 0$, and is distinct from Euler, but both have $\kappa = 1/2$. Their local difference is $O(h^3)$. On the scalar Gaussian secant family Euler is target-exact; this distinct method has $O(h^2)$ target error on the same uniform grid even though no order-elevating generator exists. Thus distinct method names alone are not sufficient either. Negative weights in this counterexample are permitted by the stated class; no strong-stability-preserving property is claimed.

a) Degenerate refinement is a genuine counterexample.: For the Gaussian secant field in (15), split each step $h = 1/N$ into h/N and $h(1 - 1/N)$. There are $2N$ fine steps, but $\beta_k = 2N^{-1}(1 - N^{-1})$. A direct second-order expansion of the two updates shows their map differs from the coarse Euler update by $O(h^3)(1 + |x|)$, uniformly in k. Consequently both target errors are $O(N^{-2})$ although no order-elevating generator exists in dimension one. Likewise, a field fitted to N uniform steps has $O(N^{-2})$ error on $N + 1$ uniform steps: their leading step sizes differ by only $O(N^{-2})$. A statement about two *distinct counts* without a nondegenerate contrast is therefore false.

b) An additional assumption for parameter distance to a generator.: The residual measures an effect of acceleration, not a distance between parameter vectors. A small residual need not place the model parameters close to those of an exact generator. That further conclusion requires local identifiability in a finite-dimensional exact-path family. Here identifiability means that every parameter direction changes the selected observations by a quantitatively nonzero amount. Let $\theta \mapsto u_\theta$ be such a family with the common derivative bound, let $R(\theta)$ collect finitely many transported Hermite coefficients at fixed times, and suppose $R(\theta_*) = 0$ for an order-elevating generator, $s_{\min}(DR(\theta_*)) \geq \sigma > 0$, and DR is L_R-Lipschitz on a parameter ball. For $\|\theta - \theta_*\| \leq \sigma/L_R$, Taylor's theorem gives

$$\|\theta - \theta_*\| \leq 2\|R(\theta)\|/\sigma.$$

The observer bounds control each coordinate of R. The proof is simply $\|R(\theta)\| \geq \sigma\|\theta - \theta_*\| - L_R\|\theta - \theta_*\|^2/2$. This is a conditional inverse-function estimate, not an assertion that an arbitrary neural parametrization is identifiable, that a nearby order-elevating generator exists, or that the selected finite tests characterize all order-elevating generators globally.

c) Why first-derivative bounds alone do not suffice.: The uniform bounds must control changes in time as well as changes in space. Otherwise, a field can oscillate between the evaluated times and fit both numerical grids without satisfying the required acceleration condition. On the scalar nonaffine Gaussian path let $a_h(t) = [g(t + h) - g(t)]/[hg(t)]$, with g smoothly extended to a neighborhood of the time strip. Define

$$b_h(t) = \frac{a_h(t - h/2)}{1 + (h/2)a_h(t - h/2)}, \tag{S61}$$

$$c_h(t) = a_h(t) + \{b_h(t) - a_h(t)\}\sin^2(\pi t/h). \tag{S62}$$

For small h, the field $u_h(t,x) = c_h(t)x$ has common bounds on $|u|/(1+|x|)$, $|\partial_t u|/(1+|x|)$, $D_x u$, $D_x\partial_t u$, and $D_x^2 u = 0$. Indeed $b_h - a_h = O(h)$ in every fixed coefficient derivative norm, so differentiation once of the oscillatory factor remains bounded. At a coarse node kh, $c_h = a_h$ and the coarse map is exactly the target scale ratio. At its midpoint, $c_h(kh + h/2) = a_h(kh)/(1 + ha_h(kh)/2)$, and the two half-step multipliers multiply to $1 + ha_h(kh)$. Both runs are therefore target-exact at *every common coarse node*. Nevertheless the path has no order-elevating generator in dimension one. The second time derivative of c_h is generally $\Theta(h^{-1})$, violating the common derivative bound precisely where compactness of the first time derivative is needed. This disproves a version of the characterization that keeps only uniform first-derivative bounds. It does not rule out replacing the second-derivative bound by an appropriate uniform modulus of continuity.

S9. Beyond Gaussian sources: a radial confinement obstruction

The source two-jet is the pair of first and second time derivatives of the distribution path at its start. The same calculation can be made for some non-Gaussian sources. To complete the obstruction proof, we must bound the squared correction perpendicular to a selected direction. We prove that bound for the radial source class below. Odd dimension alone would not supply the needed moment estimate.[9]

Let

$$q_W(x) = Z_W^{-1} \exp\{-|x|^2/2 - W(|x|^2)\}, \quad W \in C^2([0,\infty)), \quad W(0) = 0, \quad W' \geq 0. \tag{S63}$$

The normalizing constant makes the density integrate to one. The additional radial term reduces the relative weight of large radii. This includes $W(s) = \epsilon s^2/4$, $\epsilon > 0$, which is non-Gaussian. The density is positive and smooth, and its moments of every order are finite. Let $X \sim q_W$ and Y be independent, with $\mathbb{E}Y = 0$, $Y \in \mathcal{P}_2$, and covariance $S \succ 0$. Prescribe $p_t = \mathrm{Law}((1-t)X + tY)$ near zero.

Theorem S9 (Non-Gaussian source-local obstruction)**.** *If d is odd, this path has no C^1 generator near zero that is globally Lipschitz in space, whose source acceleration satisfies $a_0 \in L^1(q_W)$, and whose source acceleration preserves the source distribution: $\mathrm{div}(q_W a_0) = 0$. Every field in $\overline{\mathcal{B}}_K$ satisfies the integrability condition. Differentiation of compact flow observables twice at zero must be valid; the common derivative bound in the paper is a sufficient, but not necessary, condition. No Gaussian, symmetry, or smooth-density assumption is made on the target.*

Proof: The proof follows the Gaussian argument until the moment estimate. The first path derivative says that the correction leaves the source distribution unchanged. The second derivative relates the squared correction in a direction to acceleration in that direction. We then bound moments in the perpendicular directions for this source. With those bounds, the large-sphere argument again produces a test that detects nonzero acceleration.

Put $r = u_0 + x$ and $L = \mathrm{Lip}(r)$. Differentiating the path once gives $\mathrm{div}(q_W r) = 0$. Since r is C^1, this identity is pointwise and gives

$$\mathrm{div}\, r = (1 + 2W'(|x|^2))x \cdot r, \qquad |x \cdot r(x)| \leq dL.$$

For a unit e, the marginal flux $q_{W,e}(y)\mathbb{E}[e \cdot r(X) \mid e \cdot X = y]$ has zero distributional derivative. It is integrable and hence identically zero. The source second-jet argument consequently gives exactly

$$\mathbb{E}[\psi'(e \cdot X) e \cdot a_0(X)] = \int \psi''(y)\{e^\top S e - Q_e(y)\} q_{W,e}(y)\, dy, \quad Q_e(y) = \mathbb{E}[(e \cdot r(X))^2 \mid e \cdot X = y]. \tag{S64}$$

Indeed $(e \cdot (Y - X))^2$ has conditional mean $y^2 + e^\top S e$, while $(e \cdot u_0(X))^2$ has conditional mean $y^2 + Q_e(y)$; the cross term vanishes by the flux identity. Cutoffs, finite second moments, and $a_0 \in L^1(q_W)$ justify the extension from compact tests to the scalar bounded-derivative tests used below: the cutoff error is at most a constant times $\int_{|x|\geq R} |a_0| q_W \, \mathrm{d}x$.

Conditional on $e \cdot X = y$, the perpendicular coordinate has density proportional to $\exp(-|z|^2/2)\exp\{-W(y^2 + |z|^2)\}$. Relative to a standard Gaussian perpendicular coordinate, this density gives smaller relative weight to larger radii. The weight is nonincreasing because $W' \geq 0$. The following elementary comparison therefore bounds its increasing radial moments by the Gaussian moments. For an increasing function f and a decreasing nonnegative w, $\mathbb{E}[f(R)w(R)] \leq \mathbb{E}f(R)\mathbb{E}w(R)$: subtract using two independent copies of R and note that $(f(R) - f(R'))(w(R) - w(R')) \leq 0$. Thus the conditional moments satisfy

$$\mathbb{E}|Z_\perp|^2 \leq d - 1, \qquad \mathbb{E}|Z_\perp|^4 \leq (d-1)(d+1).$$

The same argument shows finiteness of all needed source moments.

Let $A_d = d + \sqrt{d^2 + d - 2}$ and choose $R \geq \max\{1, 2A_d L/\sqrt{\lambda_{\min}(S)}\}$. The tangential projection of r has a zero at Re for some e by the hairy-ball theorem on S^{d-1}; for $d = 1$ its tangent space is zero. For $y \in [R, R+1]$, the pointwise divergence identity and Lipschitzness give

$$|e \cdot r(ye + z)| \leq \frac{L}{R}\{d + |z|\sqrt{(y-R)^2 + |z|^2}\}.$$

[9] AI assistance for this section is disclosed in the acknowledgment [1–3].

The conditional moment bounds therefore imply $\sqrt{Q_e(y)} \le A_d L/R \le \sqrt{\lambda}/2$. A convex scalar test with $\psi'' = \mathbf{1}_{[R,R+1]}$ and $0 \le \psi' \le 1$ has positive pairing at least $(3\lambda/4)\Pr\{e \cdot X \in [R, R+1]\} > 0$ in (S64). This contradicts source acceleration preservation. An explicit positive pairing bound follows by bounding the probability below by

$$\frac{\mathrm{vol}(B_{d-1}(1))}{(2\pi)^{d/2}} \exp\left\{-\frac{(R+1)^2+1}{2} - W((R+1)^2+1)\right\}, \tag{S65}$$

using $Z_W \le (2\pi)^{d/2}$ and integrating over the perpendicular unit ball. For $d = 1$ use $\mathrm{vol}(B_0) = 1$; the displayed bound is still conservative. ∎

The two-grid characterization also allows γ_d to be replaced by a fixed positive continuous source density with a finite fourth moment. The changes to the proof are specific. In Lemmas S1–S2, use the source's second and fourth moments in place of Gaussian moment bounds. Second-moment tails justify compactness and necessity. The fourth moment controls the acceleration-flow coupling used for sufficiency, with physical time fixed. A C^1 bi-Lipschitz flow preserves the positive continuous density needed in that proof. The preceding obstruction therefore gives a uniform target-only c/N barrier for (S63) under the same common derivative bound and balanced refinements. The quantitative Hermite result does *not* extend to this non-Gaussian source without a corresponding spectral argument.

An invertible affine change of source coordinates preserves the source obstruction and Euler update algebra. If $X = A\widetilde{X} + b$, transform velocities and target covariances by A^{-1}; the derivative bound changes by the chain rule, including the affine shift and the operator norms of A, A^{-1}, and $W_2(A_{\#}\mu, A_{\#}\nu) \ge s_{\min}(A) W_2(\mu, \nu)$. This covers non-isotropic elliptical versions of the source class. It does not cover arbitrary multimodal initial densities, arbitrary heavy tails, or discontinuous velocities. It also does not cover velocities assumed only to have weak derivatives in a Sobolev space.

S10. The exact asymptotic optimum over all scalar nonlinear candidates

The characterization gives the error's order of decay, but it generally does not give the smallest coefficient multiplying that order. For the scalar Gaussian path below, we find that coefficient over *all* admissible nonlinear fields. Each field is judged by the sum of its worst coarse and refined errors over the tested times. We then choose the best field. This order of worst-time and best-field choices defines the minimax problem. No general sharp constant is established above one dimension. Let $d = T = 1$ and $p_t = \mathcal{N}(0, g(t)^2)$ with $g(t) = \sqrt{(1-t)^2 + t^2}$. Use uniform N-step Euler and its fixed m-fold refinement, $m \ge 2$. Write $q_N^{(m)} = \inf_{u \in \mathcal{B}_K}(E_c + E_r)$.[10]

Theorem S10 (Sharp all-candidate scalar constant)**.** *For any fixed derivative bound strictly exceeding* $\max\{1, \|c\|_\infty, \|c'\|_\infty, \|c''\|_\infty\}$*, where* $c = g'/g$ *on* $[0,1]$*, the secant fields belong to* $\mathcal{B}_K$ *for all sufficiently large* N*, and*

$$\lim_{N\to\infty} N q_N^{(m)} = \frac{1}{2}(1 - 1/m)(1 + \pi/2). \tag{S66}$$

In particular the bisection constant is $(2+\pi)/8$.

Proof: We first show that an asymptotically accurate sequence must approach the unique exact scalar generator. A uniform particle-error expansion then gives a lower bound on the separation between its coarse and fine numerical laws. Finally, the secant field attains that lower bound.

Any globally Lipschitz exact generator of this scalar Gaussian path is $v(t,x) = c(t)x$. Indeed the difference of two generators satisfies $\partial_x(p_t r_t) = 0$; Gaussian integrability and at most linear growth force its constant flux to be zero. This is the standard one-dimensional uniqueness argument for a generator with vanishing flux.

We first record an elementary uniform particle expansion on uniform grids. For any $u \in \overline{\mathcal{B}}_K$, with $X_t = F^u_{0,t}(G)$, let

$$B^u_t = J_t(G) \int_0^t J_s(G)^{-1} a_s(X_s)\, ds.$$

The common-start recursive Euler error satisfies, in L^2 uniformly over nodes and the derivative class,

$$\widehat{X}_k - X_{kh} = -\tfrac{h}{2} B^u_{kh} + O_{L^2}(h^2). \tag{S67}$$

To derive the expansion, write $e_k = \widehat{X}_k - X_{kh}$. Taylor's formula gives $e_{k+1} = (1 + hDu(t_k, X_{t_k}))e_k - h^2 a_{t_k}(X_{t_k})/2 + r_k$, where $|r_k| \le Ch^3(1+|G|)^2$ after using the uniform strong bound $|e_k| \le Ch(1+|G|)$ and $\|D^2 u\| \le K$. Variation of constants for $B' = Du(t, X_t)B + a(t, X_t)$ has a one-step quadrature remainder $O_{L^2}(h^2)$; multiplying it by $h/2$ and comparing the recurrences yields (S67) by discrete Gronwall. Gaussian fourth moments bound all remainder terms. This uses only the derivatives already controlled by the class.

[10] AI assistance for this section is disclosed in the acknowledgment [1–3].

Choose approximate minimizers with $E_c+E_r = O(1/N)$, possible by the secant upper construction below. Every convergent subsequence has, by the node-to-path bound and compactness, an exact limiting generator and hence limit v. Local C^1 convergence plus Gaussian tails implies $B^{u_N} \to B^v$ in uniform-in-time L^2. Subtract (S67) for steps h and h/m to obtain

$$h^{-1}(\widehat{X}^c_k - \widehat{X}^r_{mk}) \longrightarrow -\tfrac{1}{2}(1-1/m)B^v_{t_k}.$$

For $hK < 1$, each scalar Euler update is increasing. Particles therefore keep the order of their initial positions. Pairing the same initial quantiles also pairs the same output quantiles. Such a pairing is *optimal* in one dimension. Thus the W_2 distance between the two numerical laws equals the displayed common-start L^2 norm. The target triangle inequality proves the lower bound with constant $\frac{1}{2}(1-1/m)\sup_t \|B^v_t\|_2$.

For $v = cx$, direct integration gives $B^v_t = g(t)\int_0^t g''(s)/g(s)\,ds\,G$. Let $I(t) = \int_0^t 1/g(s)^4\,ds$. The product $g(t)I(t)$ is increasing. On $t \le 1/2$, $I(t) \le t/g(t)^4$, so its derivative is at least $(1-3t+4t^2)/g(t)^5 > 0$; on $t \ge 1/2$ both terms in the derivative are nonnegative. Its maximum is thus $I(1) = 1+\pi/2$, obtained by substituting $z = 2s-1$.

Finally $u_h(t,x) = [g(t+h)-g(t)]x/[hg(t)]$ converges to v in the scalar coefficient C^2 norm and satisfies the derivative bound strictly for sufficiently small h. It gives exact coarse target marginals, while its refined target error has coefficient $\frac{1}{2}(1-1/m)\sup_t g(t)I(t)$. This supplies the matching upper bound and proves the limit. The optimal constant concerns uniform grids, this fixed scalar path, and the sum-of-maxima objective; it is not a sharp constant for general dimension or optimized nonuniform grids. ∎

S11. One-dimensional geometry and the limit of endpoint tests

A quantile function gives the position below which a specified proportion of a distribution lies. For example, the middle quantile has half the probability below it. In one dimension, following each fixed quantile through time exactly transports the distribution. We can therefore read the acceleration condition directly from these quantile trajectories.[11]

Let p_t be a scalar path with a positive continuous density and a finite second moment, and let Q_t be its quantile function. Among the fields in $\overline{\mathcal{B}}_K$, the flux argument of Supplementary Section S10 shows that the unique exact generator is $u(t,Q_t(z)) = \partial_t Q_t(z)$ and $a(t,Q_t(z)) = \partial_{tt}Q_t(z)$. Thus an order-elevating generator exists exactly when $\partial_{tt}Q_t = 0$ in $L^2(0,1)$ and the resulting generator belongs to the chosen class closure. Necessity follows because the scalar weighted flux must vanish. For sufficiency, each quantile moves along a straight trajectory, so Euler reproduces that motion exactly. This is the standard one-dimensional displacement-interpolation criterion. It constructs a field for an externally specified scalar path. In several dimensions, a straight, invertible interpolation of prescribed transport maps also gives a sufficient Euler-exact construction. This construction is known in displacement geometry and is not necessary in general.

Accuracy only at the final time cannot replace accuracy along the whole path. Errors with different signs can cancel before the final time. The smooth scalar field $u(t,x) = -5t^2x$ generates $g(t) = \exp(-5t^3/3)$, and

$$\int_0^1 \{c'(s)+c(s)^2\}\,ds = \int_0^1 (-10s+25s^4)\,ds = 0.$$

The terminal Euler error is $O(h^2)$ simultaneously for every fixed integer refinement factor, while the acceleration is nonzero and the maximum-node error is first-order. Conversely, under a *specified* transported test whose pairing has one sign and satisfies $|I(t)| \ge a > 0$ on the full interval, (S46) with $i = 0, j = N$ and $Z_0 = 0$ gives a terminal c/N bound whenever the remainder is dominated. The extra sign condition is an observability assumption; it is not implied by a nonzero local residual.

S12. What the finite-sample diagnostic can establish

Theorem S7 requires population W_2 errors, or justified upper bounds on them. The sample-based sliced distances and finite-feature discrepancies do not supply those upper bounds. We therefore ask a separate statistical question: do the numerical and target distributions agree on a fixed collection of measurements?

Each measurement is the mean of a bounded test function, called a witness. The finite-witness metric is the largest absolute difference between the numerical and target means. It is a finite integral probability metric. Concentration bounds give confidence intervals for this particular metric. Simultaneous coverage means that the confidence statement applies to all reported comparisons together. Such testing and kernel mean comparisons are standard [7].[12]

Proposition S11 (Simultaneous target-fidelity intervals)**.** *Condition on all training data, fitted fields, diagnostic times, solver configurations, and a finite dictionary $f_1,\dots,f_J$ with $|f_j| \le 1$. For every numerical or target law, estimate its witness means from n_l independent draws, independent of this conditioning. Let Q_{means} be the total number of scalar means estimated. For failure probability $0 < \delta < 1$, set*

$$r_l = \sqrt{\frac{2\log(2Q_{\mathrm{means}}/\delta)}{n_l}}.$$

[11] AI assistance for this section is disclosed in the acknowledgment [1–3].

[12] AI assistance for this section is disclosed in the acknowledgment [1–3].

With probability at least $1-\delta$, simultaneously for all reported comparisons of law l and target p,

$$L_l = \max\{0, \widehat{D}_l - r_l - r_p\} \le D_{\mathcal{F}}(\mu_l, p), \tag{S68}$$

$$D_{\mathcal{F}}(\mu_l, p) \le U_l = \min\{2, \widehat{D}_l + r_l + r_p\}, \tag{S69}$$

where $D_{\mathcal{F}}(\mu,\nu) = \max_j |\mathbb{E}_\mu f_j - \mathbb{E}_\nu f_j|$ and $\widehat{D}_l$ is its empirical counterpart. Samples can be shared between resolutions, times, and target comparisons; independence across these means is not required. Independence within each sample mean is required. If each witness is also 1-Lipschitz, $L_l \le W_1(\mu_l, p) \le W_2(\mu_l, p)$.

Proof: First bound the error of each sample mean. Then add the failure probabilities to bound the chance that any one of those bounds fails. This is the union-bound step. It gives simultaneous coverage of every reported comparison and does not require different comparisons to be independent.

For a variable in $[-1,1]$, Hoeffding's bounded-mean inequality gives $\Pr(|\widehat{m} - m| > r) \le 2\exp(-nr^2/2)$. This follows, for example, by bounding the centered moment generating function by $\exp(s^2/2)$, applying Markov's inequality, and minimizing in s; the lower tail is identical. A union bound over all means gives simultaneous errors at most r_l. Triangle inequalities for each difference and then the finite maximum give (S69). The final Wasserstein inequality is the Lipschitz coupling bound. ■

Hoeffding's interval uses only the witness's range. It does not use how much the observed values vary. An empirical Bernstein interval also uses the sample variance and can therefore be narrower. For this alternative, pair each numerical sample with one independent target sample and write $D_i = f_j(\widehat{X}_i) - f_j(Y_i) \in [-2,2]$. Let v_n be its unbiased sample variance. For Q_{diff} scalar differences and $n \ge 2$, a simultaneous two-sided radius is

$$r^{\mathrm{EB}} = \sqrt{\frac{2v_n \log(4Q_{\mathrm{diff}}/\delta)}{n}} + \frac{28\log(4Q_{\mathrm{diff}}/\delta)}{3(n-1)}. \tag{S70}$$

This is the empirical Bernstein inequality of Maurer and Pontil [8, Theorem 4], scaled from $[0,1]$ to $[-2,2]$. We combine the failure probabilities for both signs and every selected difference, so that the intervals hold simultaneously. For each metric, take the maximum of $\max\{0, |\widehat{D}_j| - r_j^{\mathrm{EB}}\}$ as its lower bound and the maximum of $\min\{2, |\widehat{D}_j| + r_j^{\mathrm{EB}}\}$ as its upper bound. The experiment splits its total failure probability equally between this family and the Hoeffding family so both displays hold simultaneously.

If $L_{\mathrm{fine}} > U_{\mathrm{coarse}}$, the intervals establish worsening in this *specified finite-witness metric* at the stated confidence level. They do not establish worsening in full W_2, because U_{coarse} is not a Wasserstein upper bound. Small finite-witness discrepancy does not establish distributional fidelity. Two distributions can agree on every selected mean while differing on a test that was not included in the dictionary. A low cross-grid discrepancy alone is still weaker: the zero field has perfectly consistent outputs at every resolution while following the wrong path whenever the target moves. Neither test identifies an asymptotic order from finitely many resolutions.

The confidence statement assumes that the evaluation set does not select the models or witnesses. Selecting them on that set requires either a suitable complexity bound or a new evaluation set. A finite collection of models specified in advance is allowed if every mean is counted in Q_{means}. The experimental dictionary comprises fixed sine and cosine functions with frequency norm at most one, chosen independently of all training and evaluation samples. The intervals can be wide at 1,024 samples. An inconclusive interval establishes neither an increase nor its absence. The comparison on the larger evaluation set uses the same fitted fields with independent evaluation samples.

S13. Global regularity bounds for the tested networks

A penalty can make derivatives small at sampled inputs without bounding them elsewhere. Membership in the theoretical class requires bounds for all spatial and time inputs. We obtain sufficient bounds directly from the network's weights. These bounds can be conservative.[13]

For the two-hidden-layer tanh field $u(z) = W_3\tanh(W_2\tanh(W_1 z + b_1) + b_2) + b_3$, where $z = (x,t)$ and both hidden widths equal w, let $s_i \ge \|W_i\|_2$ and $b \ge \|b_3\|$. Since $|\tanh'| \le 1$, $|\tanh''| \le 2$,

$$\sup|u| \le s_3\sqrt{w} + b =: U, \tag{S71}$$

$$\sup\|Du\| \le s_3 s_2 s_1 =: P, \tag{S72}$$

$$\sup\|D^2 u\| \le 2s_3 s_2(s_2+1)s_1^2 =: R. \tag{S73}$$

Here derivatives are with respect to all input coordinates. The matrix norm $\|W_i\|_2$ is the spectral norm, the largest amplification of a unit input vector. The chain rule gives the first-derivative bound. For the second derivative, it gives two terms: one differentiates the outer activation twice, and the other differentiates the inner activation twice. Thus $K = \max\{1, U, P, R\}$ is a sufficient derivative bound in (1), including time derivatives; bounded functions automatically meet its growth bounds. The implementation bounds each spectral norm by the Frobenius norm, the square root of the sum of squared matrix entries. It then

[13] AI assistance for this section is disclosed in the acknowledgment [1–3].

bounds that norm by an integer using exact rational sums of the stored binary coefficients and integer square-root comparisons. Taking $\sqrt{w} \le w$ yields a fully integer, conservative certificate. This is an analytic certificate for the real-valued tanh network with those stored coefficients. It is not a rounding-error certificate for hardware inference or a certified ODE solution. These sufficient derivative bounds are too large for practically useful worst-case lower-bound constants.

S14. Complete sample-only experimental protocol

The experiments ask how target accuracy and integration accuracy change when the number of Euler steps changes. Training uses independent source and target observations, without an analytical velocity teacher. There are 18 fields in the primary experiment and 12 additional fields in the sensitivity checks, for 30 fields in total. This section gives the complete distributions, training rules, measurements, and comparison limits.[14]

A. Targets, data splits, and architecture

In dimension two, the target is a four-component Gaussian mixture with weights $(0.1, 0.2, 0.3, 0.4)$ and preliminary means $(-2,-1), (1.8,-1.5), (2,1.6), (-1.6,2)$. Subtract their weighted average from all four means. In dimension eight, expand a preliminary mean (a,b) to

$$(a, b, 0.35ab, 0.3a^2, 0.3b^2, 0.4\sin a, 0.5\cos b, 0.2(a+b)),$$

then center the eight-dimensional means. For component $k \in \{0,1,2,3\}$, the covariance is $Q_k \operatorname{diag}(s_{k,j}^2) Q_k^\top$, where $s_{k,j}$ are d equally spaced values from $0.30 + 0.04k$ to $0.70 + 0.06k$. The orthogonal matrix Q_k is the orthogonal factor in a QR decomposition of a matrix of independent, identically distributed (iid) Gaussian entries, generated with NumPy seed $90210 + 100d + k$. The code fixes the exact QR convention. In this decomposition, Q_k is orthogonal and R is upper triangular. Components have different orientations and scales; this is not a one-coordinate nonlinear product construction.

The source is standard Gaussian. The path uses independent linear interpolation, as in (9): draw the source and target endpoints independently, then join them by a line. In each run, the independent source and target training sets contain 32,768 observations each. Separate independent validation sets contain 8,192 observations each. Training minibatches independently resample indices from the source and target training sets. The training data are endpoint observations, not analytical velocities. Primary seeds are 17, 29, and 43. A field takes (x,t) as input and has two 48-unit tanh hidden layers and a linear d-dimensional output, with biases. There is no separate analytical linear field added to the network, and the requested step size is not an input. Parameter counts are 2,642 in dimension two and 3,224 in dimension eight. The fitted field is unchanged across all evaluated step counts.

B. Three objectives and fixed stopping rule

Every run uses 1,800 Adam updates, learning rate 10^{-3}, batch size 256, and gradient clipping at Euclidean norm 10. The stopping rule is fixed: training ends after that update count. Test outcomes do not select the model. Validation regression loss is measured at fixed updates but also does not select a model. For independently sampled endpoints G, Y and uniform $t \in [0,1]$, the conditional flow matching (CFM) minibatch loss is the coordinate-averaged squared error $d^{-1}|u(t,(1-t)G+tY) - (Y-G)|^2$.

The additional loss is evaluated every fourth update using 128 starting particles and independent source/target draws for the target interpolation. A fixed Gaussian frequency matrix has 64 columns, sampled with seed $155 + d$ and divided by $\sqrt{d}$. Sine and cosine yield 128 features. For their feature map Φ, set

$$L_{\mathrm{feat}}(A,B) = 128^{-1}\|\overline{\Phi}(A) - \overline{\Phi}(B)\|^2.$$

This loss compares the two sample means for each selected sine and cosine feature. It approximates a kernel mean discrepancy with finitely many features. Squaring a sample-mean difference generally adds a finite-sample variance term, so the result is a biased estimate of the squared population mean difference. Maximum mean discrepancy (MMD) compares means in the full associated kernel function space. Neither objective here is population W_2. Let L_c average the target feature loss over Euler outputs at $t = 1/4, 1/2, 3/4, 1$ with eight steps. Let L_f use sixteen steps, and let L_{cf} compare the two numerical outputs at those four times. The three training choices are CFM alone, CFM plus $30L_c$, and CFM plus $30\{(L_c + L_f)/2 + L_{cf}\}$. We call the three choices CFM, one-grid fit, and two-grid fit. The two-grid fit contains the term L_{cf} for agreement between the two runs. The optimization can stop above a population global optimum. The finite-feature objective differs from the full-W_2 error used in the inverse theorem.

[14] AI assistance for this section is disclosed in the acknowledgment [1–3].

TABLE S1
DESCRIPTIVE THREE-SEED COMPARISONS ON THE 1,024-SAMPLE EVALUATION SET. ENTRIES ARE MEANS WITH SAMPLE STANDARD DEVIATIONS IN PARENTHESES BELOW. SUBSCRIPTS GIVE EULER STEP COUNTS. LOWER VALUES ARE BETTER. SW2 USES 128 DIRECTIONS AND FIVE TIMES; RMS USES A COMMON-START NUMERICAL REFERENCE.

d	Training	$\mathrm{SW2}_8$	$\mathrm{SW2}_{16}$	RMS_8	RMS_{16}
2	CFM	0.245 (0.028)	0.197 (0.020)	0.291 (0.002)	0.147 (0.001)
2	One-grid fit	0.214 (0.039)	0.230 (0.033)	0.264 (0.003)	0.134 (0.001)
2	Two-grid fit	0.216 (0.041)	0.225 (0.036)	0.265 (0.004)	0.135 (0.002)
8	CFM	0.174 (0.015)	0.139 (0.016)	0.300 (0.008)	0.151 (0.005)
8	One-grid fit	0.148 (0.014)	0.130 (0.013)	0.243 (0.007)	0.123 (0.004)
8	Two-grid fit	0.153 (0.014)	0.133 (0.013)	0.240 (0.007)	0.122 (0.004)

C. 1,024-sample evaluation and solver evaluation counts

Evaluation uses 1,024 held-out iid numerical initial points, independent held-out source and target points for the prescribed path, and 128 fixed normalized Gaussian projection directions, with NumPy seed $701000 + d$. Each reported empirical sliced Wasserstein distance (SW2) is the square root of the mean squared difference of sorted projected samples, averaging the squared differences over both samples and the 128 directions. Equivalently, take the root mean square (RMS) of the one-dimensional Wasserstein distances along those directions. All models and solvers in a dimension use the same evaluation set and projections. This shared evaluation reduces comparison noise. It does not remove bias from the finite transport samples or error from the finite projection set. We evaluate the metric at five times, including zero. This is neither a maximum over every time node nor population full W_2.

Evaluate Euler at 4, 8, 16, 32, 64, and 128 steps. Equal-evaluation comparisons use evaluation counts 24, 48, 96, and 192 with Euler, Heun, and third-order Kutta RK3; their respective evaluation costs are one, two, and three calls per step. The numerical reference is classical RK4 with 256 steps (1,024 calls). We also compute a 512-step reference on the first 256 source points. The largest observed refinement RMS across primary models is below 2×10^{-6}. This float32 comparison does not enclose the exact error.

Own-flow error asks how closely the numerical run follows the same learned velocity. It pairs numerical and reference particles that start at the same points, then takes their root mean squared displacement. At population level, this common-start coupling with the exact flow gives an upper bound on W_2. Here the reference is numerical and the evaluation set is finite, so the reported RMS is not a certified population upper bound.

On the eight-dimensional 1,024-sample evaluation set, CFM has the lowest mean 64-step SW2, 0.1185 with sample standard deviation 0.0158. The means are 0.1323 for the one-grid fit and 0.1321 for the two-grid fit. This ordering does not hold for every seed and does not establish algorithmic superiority.

Table S3 gives the per-seed results. The irregular-grid sensitivity uses increments proportional to $1+0.8\sin(2\pi(k+1/2)/N)$, normalized to sum to one. It compares at the actual times of every $N/4$-th node, rather than at uniform quarter times. Own-flow RMS is not evaluated for these grids because its reference uses uniform diagnostic times. These grids need not be exact nested refinements; they are empirical sensitivity checks, not direct tests of the balanced-refinement theorem. On these irregular grids, all six two-dimensional one-grid and two-grid fields have lower SW2 at sixteen than at eight steps. Their SW2 then increases at 32 and 64 steps. The comparison with the uniform grids changes both the steps and the diagnostic times.

Table S2 compares the target metric and own-flow RMS at an equal count of 96 field evaluations on this 1,024-sample evaluation set. The target SW2 comparisons are descriptive for this evaluation set. The larger-set analysis evaluates only Euler and does not check their direction.

D. Sensitivity to training choices

All 12 sensitivity runs use dimension two and seed 17. They change one training choice at a time. Six use hidden widths 24 or 96 for each objective. Four use regularization weights 3 or 100 for the one-grid fit and the two-grid fit. Two add a stochastic spatial-Jacobian penalty with weight 0.01 or 0.1 to the default two-grid fit.

The Jacobian describes how the velocity changes with position. The penalty estimates its size by multiplying derivatives by a random vector whose entries independently equal -1 or 1. This is the Rademacher version of the Hutchinson estimator. Automatic differentiation evaluates it on 64 independently selected source states at random times, every fourth update. It is a local sampled penalty, not a certified global Lipschitz constraint. Each setting uses one seed, so these ablations describe sensitivity within these fitted fields. Table S4 includes all 12 settings, including those with higher target errors.

TABLE S2
EQUAL 96-FIELD-EVALUATION COMPARISONS, MEANS OF THREE SEEDS. ALL TARGET METRICS USE THE SAME HELD-OUT EVALUATION SET.

d	Training	Solver	Target SW2	Own-flow RMS
2	CFM	euler	0.18585	0.02491
2	CFM	heun	0.18763	0.00064
2	CFM	rk3	0.18766	0.00002
2	One-grid fit	euler	0.26455	0.02274
2	One-grid fit	heun	0.27325	0.00059
2	One-grid fit	rk3	0.27344	0.00002
2	Two-grid fit	euler	0.25574	0.02283
2	Two-grid fit	heun	0.26382	0.00058
2	Two-grid fit	rk3	0.26400	0.00002
8	CFM	euler	0.11673	0.02526
8	CFM	heun	0.11346	0.00048
8	CFM	rk3	0.11342	0.00001
8	One-grid fit	euler	0.13290	0.02064
8	One-grid fit	heun	0.13426	0.00037
8	One-grid fit	rk3	0.13426	0.00001
8	Two-grid fit	euler	0.13254	0.02046
8	Two-grid fit	heun	0.13363	0.00036
8	Two-grid fit	rk3	0.13362	0.00001

TABLE S3
PER-SEED 1,024-SAMPLE RESULTS FOR ALL 18 PRIMARY NEURAL FIELDS. THE UNIFORM-GRID COLUMNS GIVE SW2 AND OWN-FLOW RMS. THE IRREGULAR-GRID COLUMNS GIVE SW2 AT THEIR OWN DIAGNOSTIC TIMES, WHICH DIFFER FROM THE UNIFORM QUARTER TIMES. SUBSCRIPTS ARE EULER STEP COUNTS. LOWER VALUES ARE BETTER WITHIN EACH MEASUREMENT.

			Uniform grid				Irregular grid			
d	Training	Seed	$SW2_8$	$SW2_{16}$	RMS_8	RMS_{16}	$SW2_8$	$SW2_{16}$	$SW2_{32}$	$SW2_{64}$
2	CFM	17	0.2594	0.2090	0.2897	0.1466	0.3565	0.2419	0.2051	0.1961
2	CFM	29	0.2129	0.1736	0.2928	0.1489	0.3027	0.1971	0.1721	0.1717
2	CFM	43	0.2631	0.2082	0.2898	0.1468	0.3602	0.2417	0.2017	0.1908
2	One-grid fit	17	0.1692	0.1927	0.2677	0.1361	0.2016	0.1745	0.2008	0.2220
2	One-grid fit	29	0.2308	0.2386	0.2611	0.1333	0.2587	0.2288	0.2421	0.2559
2	One-grid fit	43	0.2421	0.2574	0.2637	0.1338	0.2714	0.2473	0.2647	0.2803
2	Two-grid fit	17	0.1701	0.1851	0.2691	0.1369	0.2134	0.1728	0.1925	0.2113
2	Two-grid fit	29	0.2326	0.2340	0.2621	0.1338	0.2681	0.2285	0.2366	0.2481
2	Two-grid fit	43	0.2465	0.2557	0.2639	0.1339	0.2829	0.2497	0.2622	0.2756
8	CFM	17	0.1582	0.1241	0.2899	0.1455	0.1951	0.1389	0.1164	0.1076
8	CFM	29	0.1768	0.1381	0.3043	0.1525	0.2185	0.1560	0.1293	0.1178
8	CFM	43	0.1880	0.1550	0.3048	0.1542	0.2269	0.1708	0.1478	0.1385
8	One-grid fit	17	0.1353	0.1188	0.2419	0.1221	0.1545	0.1237	0.1206	0.1250
8	One-grid fit	29	0.1630	0.1437	0.2363	0.1193	0.1824	0.1508	0.1393	0.1350
8	One-grid fit	43	0.1457	0.1284	0.2503	0.1266	0.1667	0.1349	0.1265	0.1309
8	Two-grid fit	17	0.1398	0.1210	0.2401	0.1213	0.1594	0.1274	0.1199	0.1238
8	Two-grid fit	29	0.1675	0.1473	0.2338	0.1181	0.1870	0.1548	0.1426	0.1378
8	Two-grid fit	43	0.1508	0.1317	0.2475	0.1253	0.1725	0.1392	0.1279	0.1292

On the 1,024-sample evaluation set, lowering the feature weight to 3 changes the 8-to-16 empirical SW2 comparison from increasing to decreasing for the tested seed. Increasing the weight to 100 gives larger 64-step estimates. The width and Jacobian changes give mixed results. On the same evaluation set, empirical SW2 increases from 8 to 16 steps for all three two-dimensional one-grid fields and all three two-grid fields, while their own-flow integration error falls. Eight-dimensional runs improve over that refinement. These are descriptive results for the evaluated fields and metric.

E. Finite-witness intervals on the larger evaluation set

The 262,144-sample evaluations were performed after inspection of the 1,024-sample results. Both evaluations use the same trained fields. Neither evaluation changes the fields or which ones are reported.

The confidence experiment evaluates all nine two-dimensional trained fields on an independent 262,144-sample evaluation set. Sixteen sine and sixteen cosine witnesses have frequencies of norm at most one. Evaluate 8, 16, and 64 Euler steps at the four nonzero quarter times. All inference here uses double precision with the fitted coefficients. Proposition S11 treats the ideal experiment with iid samples and real arithmetic. The numerical execution uses pseudorandom samples and floating-point arithmetic, whose errors are not enclosed by certified intervals.

There are $(9 \cdot 3 + 1) \cdot 4 \cdot 32 = 3584$ scalar means and $9 \cdot 3 \cdot 4 \cdot 32 = 3456$ scalar paired differences. Allocate failure probability 0.025 to the Hoeffding family and 0.025 to the empirical Bernstein family. Consequently both displayed families have joint

TABLE S4
ONE-SEED SENSITIVITY CHECKS ($d = 2$, SEED 17). THESE ARE ABLATIONS, NOT THREE-SEED EFFICACY COMPARISONS.

Setting	Training	$SW2_8$	$SW2_{16}$	$SW2_{64}$
capacity w24	One-grid fit	0.1985	0.2128	0.2434
capacity w24	CFM	0.2466	0.2047	0.1975
capacity w24	Two-grid fit	0.2017	0.2089	0.2355
capacity w96	One-grid fit	0.1899	0.2176	0.2602
capacity w96	CFM	0.2310	0.2104	0.2288
capacity w96	Two-grid fit	0.1902	0.2060	0.2429
jacobian 0.01	Two-grid fit	0.2300	0.2398	0.2665
jacobian 0.1	Two-grid fit	0.2528	0.2504	0.2670
weight 100	One-grid fit	0.2513	0.2866	0.3236
weight 100	Two-grid fit	0.2477	0.2770	0.3106
weight 3	One-grid fit	0.1697	0.1370	0.1526
weight 3	Two-grid fit	0.1737	0.1384	0.1518

TABLE S5
SLICED-DISTANCE EVALUATION ON THE LARGER EVALUATION SET IN TWO DIMENSIONS. EACH ROW IS ONE TRAINED MODEL EVALUATED ON THE SAME 262,144 HELD-OUT SAMPLES AND 128 FIXED DIRECTIONS. ENTRIES ARE MAXIMA OVER FIVE DIAGNOSTIC TIMES; LOWER IS BETTER. THESE ARE DESCRIPTIVE EMPIRICAL MEASUREMENTS.

Training	Seed	$SW2_8$	$SW2_{16}$	$SW2_{64}$
CFM	17	0.27820	0.20568	0.16570
CFM	29	0.23745	0.17249	0.14596
CFM	43	0.29447	0.22403	0.18508
One-grid fit	17	0.15552	0.14577	0.16972
One-grid fit	29	0.23913	0.22611	0.23636
One-grid fit	43	0.21501	0.20510	0.22141
Two-grid fit	17	0.16453	0.14451	0.16142
Two-grid fit	29	0.24429	0.22472	0.23007
Two-grid fit	43	0.22542	0.20909	0.22021

coverage at least 0.95 under their assumptions. Sample sharing across configurations does not invalidate the union bound. No model, time, or witness is selected after seeing the evaluation set without being included in the simultaneous family.

For the one-grid field with seed 17, the maximum finite-witness metric and its empirical Bernstein intervals are

N	$\widehat{D}$	$[L, U]$
8	0.060362	$[0.050819, 0.069906]$
16	0.051371	$[0.041676, 0.061065]$
64	0.089449	$[0.082638, 0.096260]$.

The 64-step lower bound is above the eight-step upper bound. The intervals therefore establish worsening for this finite metric at the stated simultaneous confidence level. They establish no 8-to-16 worsening. All six one-grid and two-grid fields instead have lower finite-witness point estimates at 16 steps than at 8 steps. Hoeffding intervals alone establish none of the worsening comparisons. The remaining interval comparisons are inconclusive. The finite dictionary and empirical SW2 compare different aspects of the distributions and can give different orderings. Neither metric is used as an upper bound on full population W_2.

F. Sliced distances on the larger evaluation set and feature-family check

The evaluation on the larger set uses the same 262,144 independent endpoint and initial samples as the bounded-witness diagnostic, together with the 128 projection directions from the primary evaluation. Saved coefficients are evaluated in double precision in batches of 8,192. At eight, sixteen, and 64 Euler steps, the sliced metric is evaluated at the five quarter times including zero. No model is retrained or selected. Re-evaluation checks on the 1,024-sample evaluation set differ from the stored float32 values by at most 1.6×10^{-8}; comparing double- and single-precision inference changes that metric by at most 1.7×10^{-8}. Fig. S1 shows each field at all three step counts. Table S5 gives the numerical values.

All nine sliced distances decrease from eight to sixteen steps on this evaluation set. Only the one-grid fields with seeds 17 and 43 increase from eight to 64 steps. The direction of the change from 8 to 16 steps depends on the evaluation set. The independent trigonometric witness estimates agree with the preceding diagnostic to approximately 5.1×10^{-15}. The witness metric and SW2 measure different properties of a distribution. Their values and comparisons need not agree. They are not interchangeable estimates of one metric.

To examine whether the shared sine/cosine form of training and diagnostic features explains their behavior, we compare three dictionaries on the same large evaluation set. They contain the 128 actual training features, 32 independently drawn sine/cosine features, and 32 independent functions $\tanh(v_j^\top x - c_j)$ with $|v_j| \le 1$ and $c_j \in [-2, 2]$. The last dictionary has

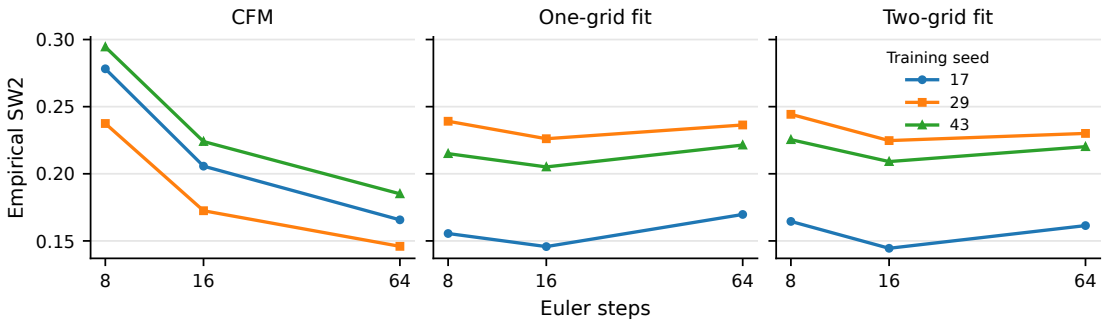


Fig. S1. Sliced Wasserstein distance (SW2) at 8, 16, and 64 Euler steps on the 262,144-sample evaluation set. Each line is one trained field. All nine distances decrease from 8 to 16 steps. At 64 steps, the one-grid fields with seeds 17 and 43 have larger distances than at 8 steps.

bounded 1-Lipschitz functions and does not use trigonometric features. We compare whether each dictionary's discrepancy increases or decreases. The dictionaries contain different numbers and types of functions, so their absolute values are not one common metric. For the six one-grid and two-grid fields, both trigonometric dictionaries decrease from eight to sixteen steps. The tanh dictionary decreases for five and increases for the one-grid field with seed 29. Thus improvement is not confined to the shared trigonometric form. This exploratory comparison neither establishes nor rules out a causal training-feature preference. It adds no simultaneous confidence claim beyond the preceding predeclared witness analysis.

G. Analytical checks and global derivative bounds

The deterministic controls check the formulas for equal refinements $m = 2, 3, 4$, the sharp scalar constant, degenerate splits, N versus $N+1$, equal-defect solver pairs, temporal aliasing, endpoint cancellation, and conditional radial moments. Numerical integration checks radial moments in selected examples. These calculations check the implementation, not the general proof. The full inverse formula uses explicit Gaussian moments and Hermite coefficients. Its floating-point evaluation is not an interval certificate, and its high-degree constants can be numerically enormous.

Every fitted network has a sufficient global derivative bound established by integer upper bounds on Frobenius norms of its exact stored binary coefficients (Supplementary Section S13). Across the 30 fitted fields, these sufficient derivative bounds range from 12,096 to 50,688. They prove class membership for the corresponding real tanh networks but are far too large for a useful application of the worst-case risk thresholds. They are not estimated local Jacobian norms or certificates for finite-precision inference.

Repetition of all 18 primary training runs in the same software environment reproduced their parameter tensors and non-timing evaluation results. Unit tests check finite computations; the analytical theorems rely on their mathematical proofs.

H. Population Wasserstein error on a three-dimensional mixture

The target in Section VIII-A is $Y = \mu\xi e_1 + \sigma Z$ with $\mu = 1.5$, $\sigma = 0.5$, symmetric sign ξ, and independent standard Gaussian Z. Use the exact conditional field (S23). In the second field, replace its two Gaussian coordinates by the linear velocity $M'(t)M(t)^{-1}x$, where $M(t) = (1-t)I_2 + \sigma t J$ and J rotates the plane by a quarter turn. This block has the same Gaussian marginals as the conditional field. Its Euler update equals $M(t+h)M(t)^{-1}x$ because M is affine in time. Both fields obey a common finite derivative bound. The mixture coordinate has the same velocity in both fields.

The exact and numerical laws each factor into a scalar law and a planar Gaussian. Any pairing must pay at least the optimal squared cost in each block. Since the blocks are independent, we can choose an optimal pairing in each block and combine them. This attains the sum of the two costs, so the squared Wasserstein distance adds across blocks. In the plane, if the numerical and target coordinate standard deviations are s and $g(t) = \sqrt{(1-t)^2 + \sigma^2 t^2}$, the squared distance is $2(s - g(t))^2$. For the skew field, this term is zero in exact arithmetic.

In the mixture coordinate, the spatial derivative of the nonlinear velocity term is nonnegative on $[0, 1]$. The spatial derivative of its linear term is bounded below by $-(1+\sigma^2)/(2\sigma) = -1.25$. All tested Euler maps are therefore increasing. Pair the numerical particle started from z with the target point at cumulative probability $\Phi(z)$, where Φ is the standard Gaussian cumulative distribution function. This pairing attains the scalar Wasserstein distance. Invert the two-component mixture distribution by bisection. Integrate the squared distance against the standard Gaussian density. Symmetry allows integration on $[0, 9]$ with twice the weight. The omitted squared tail is at most 1.07×10^{-15}. To bound it, use $|X^{\mathrm{Euler}}| \le e^{1.25}(|z| + 7.5)$ and the target quantile bound $|Q_t(\Phi(z))| \le |z| + 1.5$, then integrate the square of their sum outside $[-9, 9]$.

Doubling the Gauss–Legendre quadrature from 512 to 1,024 nodes changes each reported error sum by less than 10^{-12} relative. The largest relative residual in the quantile survival probabilities is below 4×10^{-14}. The skew block agrees with the

TABLE S6
POPULATION W_2 ERRORS FOR TWO EXACT GENERATORS OF THE THREE-DIMENSIONAL MIXTURE PATH. ERRORS ARE DETERMINISTIC QUADRATURE ESTIMATES, WITH NO TRAINING OR EVALUATION SAMPLES. LOWER ERRORS ARE BETTER. ORDER IS ESTIMATED FROM THE SUM OF THE ERRORS AT CONSECUTIVE COARSE STEP COUNTS.

	Conditional velocity			Planar skew		
N	E_c	E_f	Order	E_c	E_f	Order
8	0.1932745	0.0992033	–	0.1511015	0.0768981	–
16	0.0992033	0.0503485	0.968	0.0768981	0.0388608	0.978
32	0.0503485	0.0253755	0.982	0.0388608	0.0195436	0.987
64	0.0253755	0.0127400	0.990	0.0195436	0.0098015	0.993
128	0.0127400	0.0063833	0.995	0.0098015	0.0049083	0.996
256	0.0063833	0.0031950	0.997	0.0049083	0.0024561	0.998

exact target standard deviation within 7×10^{-16}. These are numerical checks on the quadrature and floating-point arithmetic. The tail estimate is analytical.

Table S6 gives the maxima over every shared coarse node, separately for each run. The local order is $\log_2(\mathcal{J}_{N/2}/\mathcal{J}_N)$. This example gives two observed rates under the conditions of Theorem 3. For these fields, Theorem 4 also gives an obstruction. Here $\nu = 3$, and the initial excess loss Δ_0 is zero for the conditional field and 0.5 for the skew field. Both differ from ν, so the error sum is at least of order $1/N$ for sufficiently fine grids. Both fields generate the path exactly, so the standard Euler bound gives an upper error of order $1/N$ on uniform grids. The smaller error with the planar correction concerns this product path and these two fields. The scalar optimum in Section VII concerns a different path and an optimization over fields.

S15. LEARNED SCALAR FIELDS WITH POPULATION ERRORS

This section gives the training and evaluation methods for the scalar fields of Section VIII-B. We estimate population errors by numerical integration over a Gaussian start.[15]

A. Target, fields, and training

The target is $Y = \mu\xi + \sigma Z$ with $\mu = 1.5$, $\sigma = 0.5$, a symmetric random sign ξ, and an independent standard Gaussian Z. The path is the independent interpolation (9). Thus p_t is an equal mixture of the two Gaussians $\mathcal{N}(\pm\mu t, (1-t)^2 + \sigma^2 t^2)$. The dimension is odd, so Theorem 3 applies. The exact conditional velocity is (S23) with $d = 1$.

Each field is a network with the input (x, t), two hidden layers of 48 tanh units, and a scalar output. It has 2,545 parameters. The step size is not an input. There are three objectives.

- *CFM.* The conditional flow matching loss $\mathbb{E}|u(t, (1-t)G + tY) - (Y - G)|^2$, with t uniform on $[0, 1]$ and a minibatch of 1,024 independent draws.
- *One-grid fit.* The CFM loss plus $10\, L_8$.
- *Two-grid fit.* The CFM loss plus $10\,(L_8 + L_{16})/2$.

The loss L_n is computed as follows. Run n uniform Euler steps from 1,024 new Gaussian start points. Take each of the times $1/4, 1/2, 3/4, 1$. Sort the Euler outputs and sort 1,024 new samples of p_t. Then take the mean squared difference of the sorted values. This value is the squared W_2 distance between the two empirical distributions. The loss L_n is the mean over the four times, and its gradient is taken through the Euler steps.

Each run uses 8,000 Adam updates with the learning rate 10^{-3} in double precision. New samples are drawn at each update. Each objective uses the ten seeds 0 to 9, which gives 30 fields. All settings were fixed before the evaluation, and no field was selected with the evaluation results.

B. Population errors by quadrature

Let $z \mapsto \widehat{X}_k(z)$ be the Euler map from the start point to the state at node k. We evaluate it on 8,001 equally spaced points $z \in [-9, 9]$. If the map is increasing, the pairing of equal quantiles is optimal, and

$$W_2^2(\mathrm{Law}(\widehat{X}_k), p_t) = \int \big(\widehat{X}_k(z) - Q_t(\Phi(z))\big)^2 \varphi(z)\, \mathrm{d}z.$$

Here Q_t is the quantile function of p_t, and Φ and φ are the standard Gaussian distribution function and density. We compute Q_t by bisection on the distribution function of the mixture, and the integral by the trapezoidal rule. The Gaussian mass outside $[-9, 9]$ is below 3×10^{-19}. For all 30 fields, the Euler maps increase on the quadrature nodes. We also bound the derivative of each update, $1 + \partial_x u/N$, from the network weights. For $N = 4, 8, 16$ we propagate bounds through both layers on intervals covering the real line. For $N \geq 32$ we use the sum of the negative products of weights along paths from the input to the

[15]AI assistance for this section is disclosed in the acknowledgment [1–3].

output. Each tanh derivative lies between zero and one. The smallest computed lower bound is above 0.037. These checks use double precision without directed rounding. For the exact conditional velocity, this quadrature gives the skew-field columns of Table S6, which contain only the error in the mixture coordinate, to seven digits.

The reference for the exact flow of a field is the classical fourth-order Runge–Kutta method with 1,024 steps. The *model bias* is the largest W_2 distance between this reference and the target path at 64 equally spaced times. The *integration error* at N steps compares the Euler run and the reference from the same start points. It is their largest root mean squared distance at the four shared times. The *target error* at N steps is the largest W_2 distance to the target path at the four shared times. The quantities E_c and E_f of (2) use all nodes of the coarse grid.

C. Results

Table S7 gives the results for each field. The means are in Table I of the main article.

CFM fields. The target error is smaller at 16, 64, and 256 steps than at 8 steps for all ten seeds. The mean is 0.174 at 8 steps and 0.090 at 256 steps. The model bias is between 0.023 and 0.195, with a mean of 0.091. For 3 of the ten seeds, the smallest target error is at 256 steps. For the other seeds it is at an intermediate step count, and the difference from the value at 256 steps is small.

One-grid fields. The mean target error is 0.085 at 8 steps, 0.084 at 16 steps, 0.113 at 64 steps, and 0.122 at 256 steps. The smallest target error is at 8 steps for 6 seeds and at 16 steps for 4 seeds. Compare each value with the value at 8 steps. The target error is larger at 16 steps for 6 of the ten seeds, at 64 steps for 10 seeds, and at 256 steps for 10 seeds. The exact two-sided Wilcoxon signed-rank tests for these three paired comparisons give $p = 0.695$, $p = 0.002$, and $p = 0.002$. The mean model bias is 0.125. The mean integration error decreases from 0.125 at 8 steps to 0.004 at 256 steps.

Two-grid fields. The mean target error is 0.092 at 8 steps, 0.069 at 16 steps, 0.085 at 64 steps, and 0.093 at 256 steps. The smallest target error is at 16 steps for 9 of the ten seeds. Compared with the value at 8 steps, the target error is larger at 64 steps for 4 seeds and at 256 steps for 5 seeds ($p = 0.375$ and $p = 0.846$). Compared with the value at 16 steps, it is larger at 64 steps for 10 seeds and at 256 steps for 10 seeds ($p = 0.002$ and $p = 0.002$). The same comparison with the value at 16 steps gives 10 and 10 seeds for the one-grid fields, and 0 and 0 seeds for the CFM fields. The mean model bias of the two-grid fields is 0.096.

For a fixed field, the target error at the four evaluation times approaches the bias over those times as the step count increases. The theoretical two-grid error uses all shared nodes and approaches twice the bias over the full time interval. The table gives a 64-time estimate of bias and the two-grid error at $N = 8$. Theorem 1 concerns the best error over a class of fields. A fit at small step counts decreases the target error there. For the one-grid fields it also increases the model bias. The weight of the added loss was fixed and was not tuned, and a different weight can change the size of the effect.

S16. Variability of sample-based comparisons across evaluation sets

Section S14 reports the sliced distance SW2 on one evaluation set of 1,024 samples and on one evaluation set of 262,144 samples. The two sets give different directions for the change between 8 and 16 steps. Here we measure how often the direction changes across independent evaluation sets of the same size.[16]

A. Protocol

The nine two-dimensional fields of Section S14 and its 128 projection directions are fixed. Only the evaluation samples change. One evaluation set contains start points for the numerical runs, and independent noise and data points for the target path. We draw 200 independent sets of 1,024 samples and 30 independent sets of 16,384 samples. Their random seeds are different from the seeds of those two sets. For each set we compute SW2 at 8, 16, and 64 Euler steps as the maximum over the five diagnostic times, in double precision. This implementation gives the values of Table S3 on the separate 1,024-sample set with differences below 3×10^{-8}.

B. Results

Table S8 gives the results for each field and each size. It gives the mean and the standard deviation across sets of SW2, of its change from 8 to 16 steps, and of its change from 8 to 64 steps. It also gives the percentage of sets in which SW2 increases.

Change from 8 to 16 steps. For the three CFM fields, SW2 decreases in all sets of both sizes. For the six fitted fields (one-grid fit and two-grid fit), the mean change is negative at both sizes. With 1,024 samples, the standard deviation of the change across sets is 0.014 on average over the six fields, and the mean change is -0.011. SW2 increases in 6% to 38% of the sets, as a function of the field. At least one of the six fields shows an increase in 43% of the sets, and all six fields show an increase together in 2.5% of the sets. The separate 1,024-sample set also shows all six increases. With 16,384 samples, the

[16] AI assistance for this section is disclosed in the acknowledgment [1–3].

TABLE S7
LEARNED SCALAR FIELDS: TARGET ERROR AT N EULER STEPS FOR EACH FIELD, MODEL BIAS, AND THE TWO-GRID ERROR $E_c + E_f$ OF (2) AT $N = 8$. ALL VALUES ARE POPULATION W_2 DISTANCES COMPUTED BY QUADRATURE.

Field	Seed	$N = 8$	16	64	256	Model bias	$E_c + E_f$ at $N = 8$
CFM	0	0.146	0.072	0.025	0.021	0.023	0.218
CFM	1	0.150	0.097	0.089	0.093	0.094	0.247
CFM	2	0.143	0.078	0.050	0.051	0.052	0.220
CFM	3	0.174	0.119	0.101	0.102	0.102	0.293
CFM	4	0.148	0.076	0.040	0.040	0.041	0.225
CFM	5	0.137	0.067	0.040	0.044	0.046	0.204
CFM	6	0.257	0.217	0.198	0.196	0.195	0.473
CFM	7	0.234	0.196	0.186	0.186	0.187	0.430
CFM	8	0.187	0.143	0.131	0.132	0.133	0.330
CFM	9	0.161	0.089	0.045	0.039	0.039	0.251
One-grid fit	0	0.073	0.079	0.110	0.119	0.123	0.152
One-grid fit	1	0.068	0.068	0.099	0.108	0.112	0.136
One-grid fit	2	0.088	0.090	0.114	0.122	0.125	0.178
One-grid fit	3	0.107	0.112	0.138	0.147	0.150	0.219
One-grid fit	4	0.122	0.129	0.151	0.159	0.161	0.251
One-grid fit	5	0.168	0.169	0.185	0.191	0.193	0.336
One-grid fit	6	0.065	0.041	0.068	0.078	0.084	0.106
One-grid fit	7	0.055	0.047	0.082	0.092	0.096	0.102
One-grid fit	8	0.053	0.049	0.084	0.094	0.098	0.102
One-grid fit	9	0.048	0.057	0.096	0.107	0.110	0.105
Two-grid fit	0	0.091	0.090	0.114	0.122	0.125	0.181
Two-grid fit	1	0.100	0.091	0.110	0.117	0.120	0.191
Two-grid fit	2	0.112	0.103	0.119	0.125	0.127	0.215
Two-grid fit	3	0.107	0.073	0.079	0.085	0.088	0.180
Two-grid fit	4	0.103	0.092	0.109	0.116	0.118	0.196
Two-grid fit	5	0.079	0.050	0.070	0.079	0.082	0.129
Two-grid fit	6	0.086	0.064	0.081	0.088	0.091	0.150
Two-grid fit	7	0.077	0.041	0.059	0.068	0.073	0.117
Two-grid fit	8	0.089	0.048	0.053	0.059	0.062	0.137
Two-grid fit	9	0.078	0.039	0.057	0.067	0.071	0.117

standard deviation of the change is 0.005, and SW2 increases in at most 7% of the sets. On the set of 262,144 samples, SW2 decreases for all six fields (Table S5).

Change from 8 to 64 steps. This change differs between the fields. For the one-grid field with seed 17, the mean change is positive at both sizes. SW2 increases in 74% of the sets of 1,024 samples and in 100% of the sets of 16,384 samples. It also increases on the set of 262,144 samples. This agrees with the finite-witness intervals of Section S14-E for the same field. With 16,384 samples, the values for the other two one-grid fields are 30% and 77%, and the values for the three two-grid fields are 33%, 7%, and 20%.

With these fields and directions, increases from 8 to 16 steps occur less often on sets of 16,384 samples than on sets of 1,024. The same samples are used across fields within a set, so changes for the six fields are correlated. The mean SW2 values in Table S8 are also smaller on the larger sets.

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TABLE S8

EMPIRICAL SW2 OF THE NINE TWO-DIMENSIONAL FIELDS ON INDEPENDENT EVALUATION SETS. ENTRIES ARE MEANS ACROSS SETS, WITH STANDARD DEVIATIONS ACROSS SETS IN PARENTHESES. THE TWO INCREASE COLUMNS GIVE THE PERCENTAGES OF SETS IN WHICH SW2 INCREASES FROM 8 TO 16 AND FROM 8 TO 64 STEPS.

Training	Seed	Set size	$SW2_8$	$SW2_{16}$	$SW2_{64}$	Change 8 to 16	Increase	Change 8 to 64	Increase
CFM	17	1,024	0.305 (0.029)	0.243 (0.034)	0.213 (0.040)	−0.062 (0.011)	0%	−0.092 (0.022)	0%
CFM	17	16,384	0.280 (0.006)	0.208 (0.006)	0.169 (0.008)	−0.072 (0.003)	0%	−0.111 (0.007)	0%
CFM	29	1,024	0.266 (0.042)	0.212 (0.049)	0.193 (0.053)	−0.054 (0.014)	0%	−0.073 (0.026)	0%
CFM	29	16,384	0.241 (0.012)	0.177 (0.015)	0.152 (0.018)	−0.063 (0.005)	0%	−0.089 (0.011)	0%
CFM	43	1,024	0.315 (0.042)	0.251 (0.050)	0.218 (0.056)	−0.064 (0.012)	0%	−0.097 (0.023)	0%
CFM	43	16,384	0.297 (0.010)	0.228 (0.012)	0.189 (0.015)	−0.070 (0.004)	0%	−0.108 (0.008)	0%
One-grid fit	17	1,024	0.206 (0.050)	0.201 (0.055)	0.221 (0.056)	−0.005 (0.014)	38%	+0.016 (0.023)	74%
One-grid fit	17	16,384	0.157 (0.014)	0.147 (0.015)	0.171 (0.014)	−0.010 (0.005)	7%	+0.014 (0.008)	100%
One-grid fit	29	1,024	0.260 (0.069)	0.248 (0.073)	0.259 (0.071)	−0.012 (0.012)	18%	−0.001 (0.020)	48%
One-grid fit	29	16,384	0.244 (0.018)	0.231 (0.020)	0.241 (0.020)	−0.013 (0.004)	0%	−0.003 (0.007)	30%
One-grid fit	43	1,024	0.253 (0.054)	0.246 (0.063)	0.261 (0.067)	−0.007 (0.014)	31%	+0.008 (0.022)	65%
One-grid fit	43	16,384	0.213 (0.011)	0.202 (0.015)	0.218 (0.016)	−0.011 (0.004)	0%	+0.006 (0.007)	77%
Two-grid fit	17	1,024	0.212 (0.048)	0.199 (0.054)	0.214 (0.056)	−0.013 (0.015)	20%	+0.002 (0.024)	58%
Two-grid fit	17	16,384	0.166 (0.013)	0.146 (0.015)	0.163 (0.015)	−0.020 (0.005)	0%	−0.003 (0.008)	33%
Two-grid fit	29	1,024	0.265 (0.067)	0.247 (0.072)	0.253 (0.071)	−0.018 (0.012)	6%	−0.012 (0.020)	29%
Two-grid fit	29	16,384	0.249 (0.018)	0.230 (0.020)	0.235 (0.020)	−0.019 (0.004)	0%	−0.014 (0.007)	7%
Two-grid fit	43	1,024	0.262 (0.053)	0.248 (0.063)	0.259 (0.068)	−0.013 (0.014)	20%	−0.003 (0.023)	50%
Two-grid fit	43	16,384	0.223 (0.011)	0.206 (0.014)	0.217 (0.016)	−0.017 (0.005)	0%	−0.006 (0.007)	20%

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